Anees Hussain

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Biology

Chapter wise past papers According to PMC syllabus

MDCAT21

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Biological Molecules

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Biological Molecules

1. **How many different types of amino acids are found? (1998)**
   A. 20
   B. 25
   C. 30
   D. 35

2. **Glucose is a good source of energy because (1999)**
   A. Is composed of c, n and o
   B. Has carbon linked to hydrogen?
   C. Has high energy phosphate bonds
   D. Has carbon linked to oxygen?

3. **Which of the following is not a lipid (1999)**
   A. Waxes
   B. Peptides
   C. Steroids
   D. Oils

4. **Which of the following compound is a protein (1999)**
   A. Riboflavin
   B. Adrenaline
   C. Cholesterol
   D. Hemoglobin

5. **How many kinds of nucleic acids are found in protoplasm (1999)**
   A. One
   B. Two
   C. Three
   D. Four

6. **The lipid molecule is composed of glycerol and fatty acid molecules in a ratio of (lumhs 2005)**
   A. 1:1
   B. 1:4
   C. 1:3
   D. 1:2

7. **Glycogen is a carbohydrate which is (lumhs 2006)**
   A. Monosaccharide
   B. Disaccharide
   C. Polysaccharide
   D. Glycerol

8. **The formula of glyceraldehyde is: (dmc 2005)**
   A. C3H6O4
   B. C3H6O3
   C. C6H12O6
   D. C6h14

9. **In amylopectin the linkage is: (dmc 2007)**
   A. 1,3
   B. 1,6
   C. 2,6
   D. 6,6

10. **Nitrogen is the ultimate product of : (dmc 2007)**
    A. Fats
    B. Carbohydrates
    C. Protein
    D. Organic salt

11. **Ribose and ribulose are the example of _____ Class. (kmdc 2016)**
    A. Triose
    B. Tetrose
    C. Pentose
    D. Hexose

12. **Sugar in stored in the liver as: (dmc 2000)**
    A. Starch
    B. Glycogen
    C. Maltose
    D. Cellulose

13. **Which of the following is correct: (dmc 2001)**
    A. Glucose + fructose = maltose
    B. Glucose + galactose = sucrose
    C. Glucose + glucose = maltose
    D. Glucose + glucose = sucrose

14. **Polysaccharides are synthesized in plants by the process of: (dmc 2001)**
    A. Hydrolysis
    B. Oxidation
    C. Condensation
    D. Glycolysis

15. **The bond present between carbohydrate units (dmc 2002)**
    A. Peptide bond
    B. Covalent bond
    C. Glycosidic bond
    D. Electrostatic bond

16. **An organic compound in which hydrogen and oxygen are present in the same ratio as in water is (sbbmc 2011)**
    A. Fat
    B. Carbohydrate
    C. Protein
    D. Nucleic acid
    E. All of the above

17. **Which of the following process is reverse of condensation? (sbbmc 2016)**
    A. Hydrolysis
    B. Dehydration synthesis
    C. Polymerization
    D. Vaporization

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18. Which one of the following combinations of statements is true of saccharides in living Organisms? (2015)

A . Reducing sugars and releasing water
B. Condensation of reducing sugars using water
C. Hydrolysis, releasing reducing sugars and releasing water
D. Hydrolysis, releasing reducing sugars and using water

19. Which type of protein structure contains the three dimensional structure?

A. Primary
B. Secondary
C. Tertiary
D. Quaternary
E. Both C and D

20. Steroids consist of _____ 6-membred carbon rings and one 5-membred carbon ring. (dmc 2012)

A. Four
B. Three
C. Two
D . Five
E. Six

21. Which level of protein structure maintains the helix shape of enzymes? (nts 2014)

A. Primary
B. Secondary
C. Tertiary
D. Quaternary
E. Both a & b

22. _____ is the storage form of carbohydrates in animals and human which is equivalent to the _____ in plants. (lumhs 2019)

A. Glycogen, cellulose
B. Starch, cellulose
C. Glycogen, starch
D. Starch, glycogen

23. Which of the following structural level of proteins is indicated by an association of two alpha and beta chains in the hemoglobin molecule? (jsmu 2018)

A. Primary structures
B. Secondary structures
C. Tertiary structures
D. Quaternary structures

24. Because it is capable of dissolving more substances than any other liquid, _____ is called the "universal solvent". (NTS 2019)

A. Ethane
B. Alcohol
C. Chloroform
D. Water

25. An acyl glycerol is composed of a glycerol and fatty acid molecules in a ratio of: (lumhs 2015)

A. 1:1
B. 1:4
C. 1:3
D. 1:2

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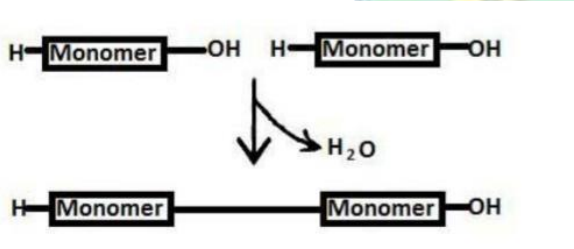
26. What occurs when sucrose is broken down to monosaccharaides? (2013)

A. Reducing sugars and releasing water
B. Condensation of reducing sugars using water
C. Hydrolysis, releasing reducing sugars and releasing water
D. Hydrolysis, releasing reducing sugars and using water

27. The structure of a lipid contains all of the following except: (nts smmbbmu 2013)

A. A carboxyl group
B. A CH_2O basic structure
C. A glycerol molecule
D. A fatty acid molecule
E. An oh group



28. The above diagram represents the process of:

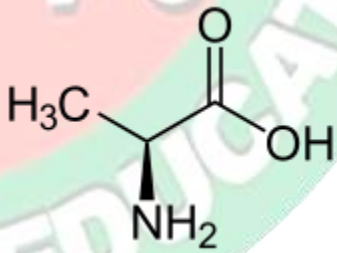
A. Hydrolysis
B. Condensation
C. Neutralization
D. Metabolism

29. Stearin (C57 H10 O6) is an important example of:

A. Carbohydrates
B. Proteins
C. Waxes
D. Acyl glycerol
E. Nucleic Acids

30. Which substance might include the above molecule?

A. Cellulose
B. Serine
C. Glucose
D. Alanine



ANSWERS KEY

1	A	6	C	11	C	16	B	21	B	26	D
2	B	7	C	12	B	17	A	22	C	27	B
3	B	8	B	13	C	18	D	23	D	28	B
4	D	9	B	14	C	19	E	24	D	29	D
5	C	10	C	15	C	20	B	25	C	30	D

Solution Video: https://youtu.be/KUw6z5iBZzc

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Past papers since 1998-2020

Enzymes

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Enzymes

1. What is the characteristic of all catalysts? (lumhs 2015)

A. They are broken down in a reaction
B. They are made of protein
C. They are not changed by the reaction
D. They cannot change the rate of reaction

2. The specificity of enzymes is due to their: (dmc 2000)

A. High m.wt
B. Ph sensitivity
C. Hydrogen bonding
D. Surface configuration
E. All of the above

3. If temperature is reduced to near or below freezing point enzymes become (baheria 2003):

A. Denatured
B. Renatured
C. Dormant
D. activate

4. When a prosthetic group joins with enzymes it forms: (dmc 2008)

A. Co-factor
B. Holoenzyme
C. Co-enzyme
D. Apo enzyme

5. Which of the following factor affect enzyme activity? (dmc 2011)

A. Temperature
B. Ph
C. Concentration of substrate
D. Radiation
E. All of the above

6. An enzyme increases the speed of a reaction: (dmc , smc 2012)

A. By adding activation energy requirements
B. By lowering activation energy requirements
C. By decreasing the concentration of products
D. By increasing the concentration of
E. All of the above

7. Some enzymes require the presence of a non-protein molecule to behave catalytically. An enzyme devoid of this molecule is called a (n) (dmc 2013)

A. Holoenzyme
B. Apoenzyme
C. Coenzyme
D. Zymoenzyme

8. The following statements are about enzymes:

1. They are globular proteins.
2. They can be inhibited by competitive inhibitors
3. They are formed in the smooth endoplasmic reticulum.
4. They are only found attached to plasma membranes in the cell.

Which statements are correct for all enzymes? (dmc , smc 2015)

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A. 1 and 4 only
B. 2 and 4 only
C. 1 and 2 only
D. 1, 2, 3 and 4

9. Which of the following is not an enzyme: (dmc 2008)
A. Polymerase
B. Hydroserae
C . Isomerase
D. Enzymate

10. When a prosthetic group joins with enzymes it forms: (dmc 2008)
A . Co-factor
B. Co-enzyme
C. Holoenzyme
D. App enzyme

11. The location of an enzyme where a substrate binds is called the: (nts 2009)
A. Binding site
B. Reaction center
C. Allosteric site
D. Lock and key model
E. Active site

12. The function of enzyme is to: (nts 2009)
A. Cause chemical reactions that would not otherwise take place
B. Change the rate of chemical reactions
C. Change the direction of reactions
D. Raise the activation energy
E. None of the above

13. The diagram shows s metabolic pathway.

$$\text{Reactant} \xrightarrow{\text{enzyme 1}} \text{Sub X} \xrightarrow{\text{enzyme 2}} \text{Sub Y} \xrightarrow{\text{enzyme 3}} \text{end product}$$

What would be the effect of adding a small amount of a non-competitive inhibitor of enzyme 2? (dmc 2013)
A. Enzyme 2 would be partially denatured.
B. Substance x would increase in concentration.
C. Substance y would no longer be formed.
D. The initial reactant would no longer be metabolized.
E. The effect would be negligible.

14. Holo enzymes consist of: (lumhs 2011)
A. Apoenzymes and prosthetic group
B. Proenzymes and prosthetic group
C. Coenzymes and prosthetic group
D. Proenzymes and apoenzymes
E. Apoenzymes and co enzymes

15. Hydrogen peroxide is highly toxic substance. In the liver it is broken down to hydrogen & oxygen by an enzyme: (lumhs 2010)
A. Amylase
B. Transferase
C. Catalase

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D. Lipases
E. Trypsin

16. The biological catalysts are called: (lumhs 2012)

A. Proteins
B. Carbohydrates
C. Vitamins
D. Minerals
E. Enzymes

17. Which of the following statements about enzyme kinetics is false? (nts smmbbmu 2013)

A. An increase in the substrate concentration (at constant enzyme concentration) leads to proportional increase in the rate of reaction.
B. Most enzymes operating in the human body work best at a temperature of 37c
C. An enzyme is generally inactivated rapidly by exposure to ultraviolet light.
D. Competitive inhibitors reduce the productivity of enzymes by blocking the substrate from entering active site.
E. None of the above

18. In Acetyl Coenzyme (Acetyl coa) the Coenzyme A, a Sulphur containing compound is derived from:

A. Vitamin A
B. Vitamin D
C. Vitamin B
D. Minerals
E. Proteins

19. A _____ is mostly a non-protein chemical compound that is required for the protein's biological activity. (LUMHS 2019)

A. Active site
B. Substrate
C. Cofactor
D. Enzyme

20. The following reaction occurs in the human gastro-intestinal tract. (liyari 2016, dmc 2013)

What are the catalyst and the product?

	Catalyst	**product**
A.	amylase	glucose
B.	trypsin	amino acid
C.	Amylase	maltose
D.	Lipase	amino acid

ANSWERS KEY

1	c	6	B	11	A	16	E
2	D	7	B	12	B	17	E
3	C	8	C	13	A	18	C
4	B	9	D	14	A	19	C
5	E	10	C	15	C	20	C

Solution Video: https://youtu.be/S0ytgyRjIlY

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Past papers since 1998-2020

Cellular structure and functions.

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Cell structure and Functions

1. The nucleus of a living cell was discovered by (1999)

A. Robert hooke
B. G. Mendel
C. Galileo
D. Robert brown

2. Which of the following is absent in plant cell? (1999)

A. Centriole
B. Centromere
C. Lysosome
D. Leucoplast

3. A dense, deeply staining, almost spherical body embedded in the cytoplasm is called: (1999)

A. Plastid
B. Golgi bodies
C. Nucleus
D. Endoplasmic reticulum

4. Ribosomes are attached to (lumhs 2005, dmc 2005, pumhs 2015)

A. Lysosomes
B. Each other
C. tRNA
D. Endoplasmic reticulum

5. The main structure inside the nucleus composed of DNA and protein and visible distinctly during cell division are

A. Chromosomes
B. Chromatids
C. Chromatin
D. Nucleolus

6. Chloroplast contain substational amount of _____ and are capable of programming synthesis of some of their new structural components: (baheria 2004)

A. Chlorophyll
B. DNA
C. RNA
D. Chlorophyll a

7. The % of lipids in protoplasm is about: (baheria 2004)

A. 2 to 3%
B. 2 to 4%
C. 2 to 5%
D. 10%

8. Leucoplast is: (baheria 2004)

A. Found in underground parts of the plant
B. Found in aerial parts of plants
C. They have chloroplast
D. They are found exogenously

9. Membrane bound organelles include all of the following, except: (sbbmc 2011)

A. Mitochondria

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B. Lysosymes
C. Peroxisomes
D. Centriole
E. Plastid

10. Rough endoplasmic reticulum are involved in: (dmc 2006)

A. Protein synthesis
B. DNA synthesis
C. RNA synthesis
D. Protein modification

11. Many ribosomes attached to endoplasmic reticulum form?(dmc 2006)

A. Abnormal function
B. Polyribosomes
C. Intense protein synthesis
D. Cell becomes malignant

12. Mitochondria can be found in: (dmc 2006)

A. Virus
B . Bacteria
C. Protozoan
D. Nostoc
E. None of the above

13. Protoplasm was first observed by:

A. Purkinje
B. Dujardin
C. Robert hook
D. Von mohl

14. The _____ structure was first proposed by robertson: (dmc 2006)

A. Cellulo-protein
B. Star protein
C. Cylo protein
D. Lipo protein

15. Outer membrane of mitochondrion and endoplasmic reticulum are similar in having: (dmc 2006)

A. Similar structure and constituents' substance
B. The enzyme monoamine oxidase
C. The enzyme nadh cytochrome reductase
D. Identical metabolic function

16. Which of the following contains enzymes for the detoxification of alcohol? (kmdc 2016)

A . Ribosomes
B. Peroxisomes
C. Glyoxysomes
D. Lysosomes

17. The rough endoplasmic reticulum differs from the smooth endoplasmic reticulum due to the presence of (lumhs 2011)

A. Lysosomes
B. Ribosomes
C. Mitochondria
D. Golgi apparatus

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E. Histones

18. In which of the following cell organelles (the structures that are site of protein synthesis) can you find high concentration of RNA? (lumhs 2015)

1. *Endoplasmic reticulum*
2. *Golgi apparatus*
3. *Ribosomes*
4. *Mitochondria*

A. 1 only
B. 2 only
C. 3 only
D. 1 and 3 only
E. 2 and 3

19. Tay-sachs disease occurs when cells are unable to produce an enzyme, leading to build up of certain lipids in cells. Which cell structure would not function correctly, resulting in the disease? (lumhs 2015)

A. Golgi apparatus
B. Lysosome
C. Mitochondrion
D. Smooth endoplasmic reticulum

20. Which one of the following possesses the most prominent golgi apparatus? (dmc 2001)

A. An islet of Langerhans
B. A retinal rod cell
C. A striated muscle
D. A xylem tracheid
E. An erythrocyte

21. Which one of the following is a characteristic of mitochondria? (dmc 2001).

A. They store proteolysis enzymes
B. They provide sites for anaerobic respiration
C. They produce cellular secretions
D. They release energy via Krebs cycle
E. They produce glucose

22. In folding's in the mitochondrial inner membrane is called: (dmc 2002)

A. Cristae
B. Thylakoid
C. Glyoxysome
D. Chromatic network

23. The nonliving part of protoplasm is: (dmc 2002)

A. Golgi bodies
B. Inclusion
C. Organelles
D. E.R

24. The movement of cytoplasm is called: (dmc 2002)

A. Exocytosis
B. Endocytosis
C. Pinocytosis
D. Cyclosis

25. Digestion of cell contents by lysosomes is termed as _____.

A. Endocytosis

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B. Autophagy
C. Immunolysis
D. Exocytosis

26. When solid particles are ingested by cell through plasma membrane: (baheria 2004)

A. Pinocytosis
B. Phagocytosis
C. Endosmosis
D. Exomosis

27. In plant cell, cell wall and cell plate is the product of: (dmc 2008)

A. Golgi bodies
B. Mitochondria
C. Endoplasmic reticulum
D. Plastids

28. According to _____, protein layers are not confined to the surface of the membrane but embedded in lipid layers. (dmc 2009)

A. Lock and key mode
B. Induce fit model
C. Fluid mosaic model
D. Lotka volterra model
E. All of the above

29. Lysosomes function in: (dmc 2011)

A. Protein synthesis
B. Processing
C. Intracellular digestion
D. Lipid synthesis
E. Carbohydrate synthesis

30. Lysosomes are also called: (dmc, smc 2012)

A. Suicide sacs
B. Chondriosome
C. Storage organelle
D. Dictyosome
E. Power house of a cell

31. The following are all features of eukaryotic cells:

I. Chloroplast
II. Ii. Peroxisome
III. Iii. Lysosome
IV. Iv. Mitochondrion
V. V. Nucleus

Which of these have a double membrane? (dmc, smc 2014)

A. I, ii and iv
B. I, iii and v
C. I, iv and v
D. Ii, iii and v
E. Ii, iii and iv

32. Which of the following statements concerning nucleolus is correct? (dmc ,smc 2014)

A. It disappears at the time of cell division
B. There is only one nucleolus in every cell
C. It plays important role in the synthesis of ribonucleic acid and ribosomes in prokaryotic cells

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D. It helps in destroying worn out organelles
E. It captures energy for the cell.

33. It is the non-living component of cell, it is secreted and maintained by the living portion of cell, and is found only in plants. What is this structure? (sbbmc 2015)

A. Golgi apparatus
B. Lysosome
C. Mitochondrion
D. Cell wall

34. The process of membrane fusion and the movement of material out of the cell, is called: (sbbmc 2016)

A. Phagocytosis
B. Endocytosis
C. Pinocytosis
D. Exocytosis

35. Which of the following nutrients is incorrectly paired with its function in a cell? (sbbmc 2016)

A. Potassium formation of cell-wall.
B. Magnesium constituent of chlorophyll.
C. Nitrogen -constituent of protein.
D. Phosphorous component of nucleic acid.

36. It is the non-living component of cell, it is secreted and maintained by the living portion of cell, and is found only in plants. What is this structure? (sbbmc 2015)

A. Golgi apparatus
B. Lysosome
C. Mitochondrion
D. Cell wall
E. Ribosome

37. The statements are all descriptions of cell structures.

1. *Surrounded by a single membrane and Enclosing a large fluid-filled space*
2. *Surrounded by a single membrane and enclosing digestive enzymes*
3. *Formed by two membranes enclosing a Matrix, the inner membrane is folded*
4. *Formed by a membrane that has flattened sacs and tubular structures inter-connected throughout the cell*
5. *Formed of nucleic acid and protein*

Which row shows the typical cell in which these cell structures are found? (dmc, smc 2015)

	Plant cell	Animal cell
A.	1, 3, 4 and 5	2, 3, 4 and 5
B.	1, 2, 3 and 4	1, 2, 3 and 5
C.	2, 4, and 5	1, 4 and 5
D.	3, 4 and 5	2, 3 and 5

38. When a red stain is added to a culture containing both living and dead cells, only the dead cells take up the stain.
Which structure (s) prevent (s) the stain entering the living cells? (dmc 2014)

A. Cell membrane
B. Cell wall
C. Cytoplasm
D. Vacuole
E. All of the above

39. Which of the following is not a difference that would allow one to distinguish between a prokaryotic and a eukaryotic cell?

I. *Presence or absence of the nucleus*
II. *Presence or absence of the cell wall*

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III. *Membrane-bound versus no membrane-bound organelles* **(dmc 2013)**

A. I only
B. Ii only
C. Iii only
D. I and ii only
E. I, ii and iii

40. Of the following, which is the incorrectly Paired one? (dmc 2013)

A. Robert hooke ... Cell wall
B. Schleiden and schwann … cell theory
C. Robert brown … nucleus
D. Watson and crick dna model

41. The food reserved co2 and h2o are stored in: (dmc 2008)

A. Plastids
B. Mitochondria
C. Ribosomes
D. Nucleus

42. Which structures contain a cell nucleus?

Key:

✓ = nucleus

✖ = no nucleus

	Red blood cells	**Root hair cell**	**Xylem vessel**
A.	✓	✓	✓
B.	✓	✖	✖
C.	✖	✖	✓
D.	✖	✓	✖

43. Which of the following is not an organelle? (lumhs 2007)

A. Nucleus
B. Golgi bodies
C. Lysosome
D. Chlorophyll
E. Chloroplast

44. Which of the following is a correct association? (lumhs 2007)

A. Mitochondria: transport materials from the nucleus to the cytoplasm
B. Lysosome: digestive enzymes for intracellular use
C. Endoplasmic reticulum: selective barrier for the cell
D. Ribosome: electron transport chain

45. Which of the following is not the function of cell membrane? (jsmu 2018)

A. Protection of cytoplasm
B. Regulating the passage of different molecules
C. Protein synthesis

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D. Cellular communication
E. Cellular transportation

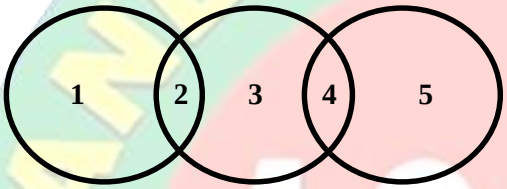
46. Which one of the following organelles is required for aerobic respirations? (jsmu 2018)

A. Nucleus
B. Mitochondria
C. Endoplasmic reticulum
D. Plastids
E. Cytoskeletons

47. A group of biologically active molecules formed from amino acids which interact with the surface of the lipid bilayer of cell membranes are called _____. (lumhs 2019)

A. Integral proteins
B. Peripheral proteins
C. Cell wall
D. Plasmodesmata

48. The diagram shows some similarities between golgi apparatus mitochondria and suicide sacs. (dmc 2015)



	1	2	3	4	5
A	Golgi apparatus	enzymes	mitochondria	Membrane bound	Suicide sacs
B	Golgi apparatus	enzymes	mitochondria	Non membrane bound	Suicide sacs
C	Suicide sacs	Non membrane bound	Golgi apparatus	enzymes	mitochondria
D	Suicide sacs	Membrane bound	mitochondria	Membrane bound	Golgi apparatus

49. Which of the following contains digestive enzymes? (lumhs 2016)

A. Mitochondria
B. Endoplasmic reticulum
C. Golgi apparatus
D. Lysosome

50. The cell membrane is made up of: (lumhs 2010)

A. Peptidoglycan
B. Lipids & proteins
C. Carbohydrates
D. Proteins & sugar
E. Cellulose

51. The chromosomes with equal arms resembling the letter 'v' are called: (lumhs 2012)

A. Telocentric
B. Dub-metacentric
C. Metacentric
D. Acrocentric
E. Autosomes

52. The diagram below represents the structure of a chloroplast as seen in section. Which label, a, b, c, d or e, indicates the site of the light reaction of photosynthesis? (lumhs 2014)

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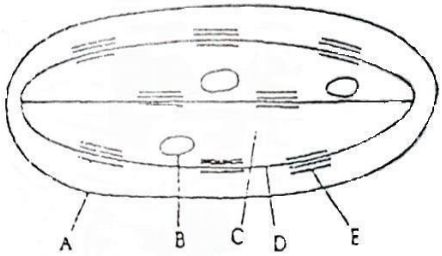

A. A
B B.
C. C
D. D
E. E

53. Which of the following is not a function of the endoplasmic reticulum? (nts smmbbmu 2013)

A. Lipid synthesis
B. Poison detoxification
C. Protein synthesis
D. Aerobic respiration
E. Storage and release of ca+2 ions

54. Which of the following is the correctly matched? (NTS 2017)

A.	**Ribosomes**	**F.**	**Detoxification of alcohol**
B.	**Lysosomes**	**G.**	**Formation of astral ray**
C.	**Centrioles**	**H.**	**Protein synthesis**
D.	**Peroxisomes**	**I.**	**Destroyers of protein particles**
E.	**Smooth ER**	**J.**	**Converts cholesterol into vitamin D in skin**

A. A=H, B=I, C=G, D=F, E=J
B. A=J, B=I, C=G, D=F, E=H
C. A=H, B=J, C=G, D=F, E=I
D. A=H, B=I, C=J, D=F, E=G

55. Which of the following statement(s) is/are correct regarding nucleus? (NTS 2017)

I. Stores waste and other substances
II. Contains genetic material
III. Helps in cellular transportation system
IV. Control center of the cell

A. I only
B. I & II only
C. II & III
D. II & IV
E. III & IV

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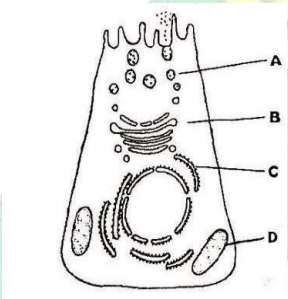
56. What features do animal cells share with plant cells? (NTS 2017)

	Chloroplast	Cytoplasm	Nucleus	Mitochondria
A .	√	√	√	√
B .	√	×	√	×
C .	√	×	×	×
D .	×	√	√	√

57. The diagram is taken from an electron micro graph of a cell, name the organelle labeled B: (liyari 2016, 2013)



A. Nucleus
B. Golgi complex
C. Lysosome
D. Mitochondrion
E. Vacuole

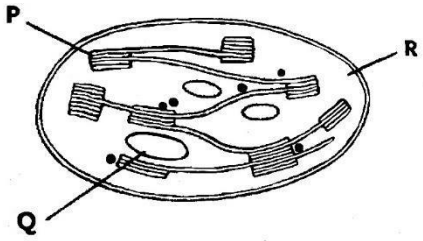
58. Which process allows the movement of molecules that are too large to enter through a cell surface membrane? (Liyari 2013)

A. Active transport
B. Endocytosis
C. Exocytosis
D. Facilitated diffusion

59. An animal cell and plant cell are placed in the distilled water. The animal cell sells and bursts, while the plant cell swells and does not bursts. What account for this difference? (Liyari 2013)

A. Animal cell have no cell wall
B. Animal cell have no vacuole
C. Plant cell surface membranes are partially permeable
D. Plant cell walls are freely permeable.

60. The diagram shows the ultra-structure of a chloroplast as seen in section. What are the functions of P, Q and R?



	P	Q	R
A.	Light reaction	Carbohydrate synthesis	Carbohydrate storage

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B.	Light reaction	Carbohydrate storage	Carbohydrate synthesis
C.	Light reaction	Carbohydrate synthesis	Carbohydrate storage
D.	Carbohydrate storage	Carbohydrate synthesis	Light reaction

61. Cell permeability and transport process of cell membrane depend upon its _____ component. (LUMHS 2019)

A. Phospholipid
B. Carbohydrates
C. Polysaccharide
D. Cellulose

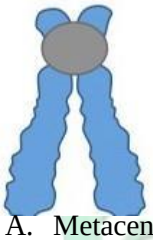
62. When plant cell receives the signal for Cell **death, it commits suicide by rupturing: (LUMHS 2019)**

A. Nucleus
B. Cell membrane
C. Tonoplast
D. Chloroplast

63. In egg, Ca^{+2} plays important role especially at the time of fertilization. Therefore, _____ are present in may thousand number inside the cell membrane. (LUMHS 2019)

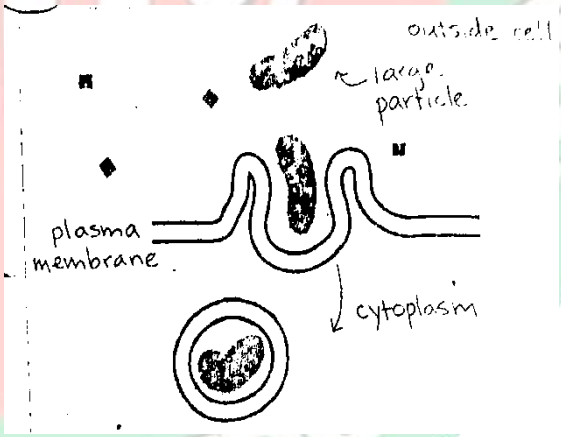
A. Ribosomes
B. Chloroplast
C. Mitochondria
D. Endoplasmic reticulum

64. Name the type of chromosome: (NTS 2019)



A. Metacentric
B. Sub meta centric
C. Telocentric
D. Sub telocentric

64. Label the following phenomenon. (NTS 2019)



A. Exocytosis
B. Osmosis
C. Diffusion
D. Phagocytosis

ANSWERS KEY

1	D	14	D	27	A	40	A	53	D
2	A	15	A	28	C	41	A	54	A
3	C	16	B	29	C	42	D	55	D
4	D	17	B	30	A	43	D	56	D
5	A	18	C	31	C	44	B	57	B
6	B	19	A	32	A	45	C	58	D
7	A	20	A	33	D	46	B	59	A
8	A	21	D	34	D	47	B	60	B
9	D	22	A	35	A	48	D	61	A
10	A	23	B	36	D	49	D	62	C
11	B	24	D	37	A	50	B	63	D
12	C	25	B	38	A	51	C	64	D
13	B	26	B	39	B	52	E	65	D

Solution Video: https://youtu.be/WEx1sgWxSDs

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Acellular Life

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Acellular Live (Biodiversity)

1. In 1892 a Russian biologist _____ demonstrated that a virus could be transmitted from an infected organism to a healthy organism of the same kind (1998)

A. Darwin
B. Iwanowsky
C. T.H Morgan
D. Lanark

2. The nucleic acid of nearly all bacteriophage is (1998)

A. RNA
B. Protein
C. DNA
D. None of these

3. The protein coat of virus is known as (1999)

A. Plasmid
B. Capsule
C. Plastid
D. Capsid

4. A virus is considered a parasite it

I. Harms it's host
II. It. Kills it's host
III. it Cannot reproduce outside it's host

A. I only
B. Il only
C. I and ll only
D. Il and lll only
E. I, ll and lll

5. Encircle the miss-match:

A. Colds... Rhino virus
B. Yellow-fever... Bacteria
C. Encephalitis... Virus
D. Minus strand RNA virus... Paramylons virus
E. Rabies... Dog bite

6. Five steps of the lytic cycle of a t4 phage are given below:

1. *The t 4 phage uses its tail to stick to specific receptor*
2. *The cell's metabolic machinery, directed by phage dna, produces phage proteins and nucleotides from the cell's degraded dna, are used to make copies of the phage genome.*
3. *The phage then directs production of an enzyme that digest the bacterial cell wall.*
4. *The empty capsid of the phage is left as a ghost outside the cell.*
5. *The sheath of the tail contracts, thrusting a hollow core through the wall and membrane of the cell.*

Which of the following is the correct sequence of the cycle?

A. 3 → 5 → 4 → 2 →1
B. 2 → 4 → 5 → 1→ 3
C. 5 → 1 → 4 → 2→ 3
D. 1 → 5 → 4 → 2 →3
E. 4 → 5 → 1 → 3 →2

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7. Chicken pox is caused by: (dmc 2011)
A. Hepatitis a virus
B. Varicella zoster virus
C. Influenza virus
D. Human immunodeficiency virus
E. Rabies virus

8. The viruses are: (dmc 2011)
A. Cellular
B. Prokaryotes
C. Non-cellular
D. Eukaryotes
E. Visible with naked eye

9. A bacteriophage consists solely of: (dmc 2011)
A. Dna and protein
B. Rna and protein
C. Rna only

10. All of the following are symptoms of hepatitis, except: (dmc 2007)
A. Deficit immunity
B. Enlargement of liver
C. Fever and fatigue
D. Jaundice

11. The TMV first appeared on the leaves of tobacco plant and was characterized by: (dmc 2007)
A. Wrinkling of leaves
B. Yellow patches on lamina surface
C. Falling off of the leaves
D. Transmission from one organism to another

12. All Viruses: (lumhs 2007, NTS 2017, liyari 2013)
A. Carry DNA
B. Carry RNA
C. Lack protein
D. Cannot reproduce outside the cell

13. Which of the following is not the viral disease? (jsmu **2018)**
A. Aids
B. Malaria
C. Influenza
D. Chicken pox
E. Rabies

14. Which one of the following is not a mode of transmission of aids? (lumhs 2019)
A. Through unsterilized needles
B. Through contact with open wounds
C. Through blood transfusion
D. Through holding hands

15. All of the following diseases are caused by viruses except: (lumhs 2010)
A. Poliomyelitis
B. Common cold
C. Yellow fever

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D. Diphtheria
E. Encephalitis

16. A viral reproductive cycle that culminates in death of host cell is known as: (lumhs 2012)
A. Lysogenic cycle
B. Lytic cycle
C. Krebs's cycle
D. Glycolysis
E. Electron-transport chain

17. Viruses resemble living things because they: (nts 2017)
A. Circulate
B. Move
C. Reproduces
D. Are crystalline
E. Are able to respond to stimuli in the environment

18. A patient is suffering from a disease s (he) has following symptoms:
I. Abdominal pain
II. Jaundice
III. Liver enlargement
IV. Fatigue
S (he) is probably suffering from (dmc 2010)
A. HIV
B. Rabies
C. Hepatitis
D. Septicemia
E. Kaposi's sarcoma

19. Genome of virus is composed of _____(LUMHS 2019)
A. DNA
B. RNA
C. Both A & B
D. Protein

ANSWERS KEY

1	B	6	D	11	D	16	B
2	D	7	C	12	D	17	C
3	D	8	C	13	B	18	C
4	C	9	A	14	D	19	C
5	B	10	A	15	D		

Solution Video: https://youtu.be/ZggTtRcAngI

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Past papers since 1998-2020

Kingdom Prokaryota

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Kingdom prokaryota

1. The main feature of cell of bacteria and cynobacteria, which distinguishes them from other organisms, is the absence of distinct of distinct membrane bounded (1998)

A. Cytoplasm
B. Dna
C. Nucleus
D. Chromosomes
C. Protein

2. _____ are rod shaped. (1998)

A. Bacilli
B. Cocci
C. Spiral
D. None of these

3. Which of the following is not a bacterial disease? (dmc 2006)

A. Typhoid and t b
B. Rabies and polio
C. Diphtheria and cholera
D. Dysentery and diarrhea

4. which of the following structure (s) is/are found in a generalized bacterial cell? (dmc 2014)

A. Flagellum
B. Pili
C. Capsule
D. Cell wall
E. All of the above

5. The cell wall of most of the bacteria have unique macromolecule called _____.

A. Cellulose
B. Chitin
C. Fibres
D. Fats
E. Peptidoglycan

6. Bacterial pili help in: (dmc 2011)

A. Locomotion
B. Conjugation
C. Phagocytosis
D. Pinocytosis
E. Exocytosis

7. Which structure in bacteria composed of polysaccharides and lipids that prevents bacteria from antibiotics and increases its pathogenicity? (dmc 2007)

A. Flagella
B. Capsule
C. Cell membrane
D. Cytoskeleton

8. Photosynthetic bacteria use which of the following instead of h2o? (dmc 2007)

A. H2S
B. H2SO4
C. H2 gas
D. Atomic hydrogen

9. Which of the following is a bacterial disease: (dmc 2007)

A. Leishmaniasis
B. Tetanus
C. Measles

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D. Polio

10. The photogenic bacteria transfer DNA material to another bacteria by: (dmc 2008)

A. Transduction
B. Transformation
C. Conjugation
D. Fragmentation

11. Vibrio. Cholera is an example of _____ type of bacteria. (lumhs 2016)

A. Cocci
B. Bacilli
C. Spirilla
D. Commas

12. The hollow, filamentous appendages which help in conjugation in bacteria are: (lumhs 2010)

A. Flagella
B. Cell wall
C. Capsule
D. Pilli
E. Storage bodies

13. Almost all the prokaryotes belong to the kingdom: (lumhs 2012)

A. Fungi
B. Plantae
C. Animalia
D. Protista
E. Monera

14. Bacilli bacteria are _____ shaped. (lumhs 2012)

A. Spherical
B. Spring
C. Rod
D. Screw
E. Comma

15. A chain of Coccus is seen in bacteria called: (Nts smmbbmu 2013)

A. Micrococcus
B. Diplococcus
C. Streptococcus
D. Sarcinia
E. Staphylococcus

16. When the tuft of flagella are present at both the ends in the structure of bacterial cell, then the condition is known as: (NTS 2017)

A. Atrichous
B. Lophotrichous
C. Peritrichous
D. Amphitrichous
E. Bitrichous

ANSWERS KEY

1	C	5	E	9	B	13	E
2	A	6	B	10	C	14	C
3	B	7	B	11	D	15	C
4	D	8	A	12	D	16	D

Solution video: https://youtu.be/ZggTtRcAngI

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Diversity among animals

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Diversity among animals

1. Skin gland of chondrichthyes secrets large amount of (1999)
A. Water
B. Sebum
C. Sweet
D. Mucus

2. Earth worm moves by using its (1999)
A. Bones and muscles
B. Setae and muscles
C. Cilia and muscles
D. Skin and muscles

3. Kangaroo belongs to (1999)
A. Prototheria
B. Eutheria
C. Metatheria
D. None of these

4. Which one of the following is not motile (2000)
A. Hydra
B. Jelly fish
C. Obelia
D. None of these

5. Sessile animals get their food by (2000)
A. Photosynthesis
B. Water filtration
C. Phagocytes
D. All of these

6. Bilaterally symmetrical animals are _____ (2000)
A. Animals which cannot be divided into two symmetries
B. Animals which can be divided into two equal parts
C. Animals which cannot be divided into two equal halves from lateral side
D. All of these

7. Echinidermates are (2000)
A. Asymmetrical
B. Bilateral symmetrical
C. Radial symmetrical
D. All of these

8. Which of the following is exclusively marine (lumhs 2005, SBBMC 2015)
A. Mollusca
B. Echinidermata
C. Annelida
D. Arthropod

9. Which organism is the least likely related to the others? (lumhs 2006)
A. Lizard
B. Frog
C. Alligator
D. Turtle
E. Snake

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10. Body cavity in arthropods is called? (Lumhs 2006)
A. Coelom
B. Acoelom
C. Pseudocoelom
D. Haemocoelom

11. Echinoderms have a brain: (dmc 2006)
A. Well-developed
B. Rudimentary
C. Absent
D. None of the above

12. Male honey bees are (dmc 2006)
A. Diploid
B. Haploid
C. Triploid
D. None

13. Pheretima posthuma is the scientific name of: (lumhs 2015)
A. Planaria
B. Liver fluke
C. Tape worm
D. Ascaris
E. Earth worm

14. Group Deuterostomata includes phylum (kmdc 2016)
A. Chordata
B. Annelida
C. Arthropoda
D. Echinodermata
E. Both A and D

15. Which of the following are animals with pointed ends and a tube like digestive track? (Dmc, SMC 2016)
A. Annelida
B. Arthropoda
C. Nematode
D. Mollusca

16. Molluscus have an exoskeleton made up of: (dmc 2016)
A. Protein
B. Silica
C. Caco3
D. Cuticles

17. Example of cnidarians includes all except: (lumhs 2009)
A.corals
B.jellyfish
C.hydra
D.sponges
E.seaanemone

18. Class Elasmobranch have an exoskeleton of
A. Placoid scales
B. Cycloid scales
C. Ctenoid scales
D. Epidermal scales

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19. What is the correct order of arthropod groups, from those with most legs to those with fewest legs? (dmc 2014)
A. Arachnids… crustaceans… insects… myriapods
B. Crustaceans… myriapods... Insects… arachnids
C. Insects... Arachnids... Myriapods... Crustaceans
D. Myriapods… arachnids… crustaceans… insects
E. Myriapods... Crustaceans... Arachnids… insects

20. Identify the phylum in which the larva is bilaterally symmetrical but the adult is radially symmetrical: (dmc 2013)
A. Ctenophora
B. Coelenterate
C. Echinodermata
D. Sipunculoidea

21. Archaeopteryx is a transitional stage between the members of which one of the following pairs? (dmc 2013)
A. Amphibian bird
B. Fish... Amphibian
C. Reptile... Mammal
D. Reptile… bird
E. Mammal... Man

22. Which vertebrate groups have scaly skin? (dmc 2013)
A. Amphibians and fish
B. Amphibians and mammals
C. Fish and mammals
D. Fish and reptiles

23. Male and female sea urchins release their sperm and eggs into the water where fertilization takes place. How can their reproduction be described? (dmc 2013)
A. Asexual reproduction which results in genetically dissimilar offspring
B. Asexual reproduction which results in genetically identical offspring
C. Sexual reproduction which results in genetically dissimilar offspring

24. Which one of the following is fish? (dmc 2012)
A. Star fish
B. Jelly fish
C. Cattle fish
D. Sea horse
E. None of the above

25. Species of phylum Platyhelminthes are: (dmc 2011)
A. Round worms
B. Flatworms
C. Hookworms
D. Threadworms
E. Pin worms

26. Characteristics features of Echinoderm is:(2011)
A. Canal system
B. Water vascular system
C. Tracheal system
D. Blood vascular system
E. None of the above

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27. _____ is commonly known as Hook worm: (dmc 2009)
A. Ancylostoma duodenale
B. Ascaris lumbricoides
C. Enterobius vermicularis
D. Hirudinaria
E. Wuchereria

28. An organism appears to be a segmented worm. Upon observation it is determined that the organism has a closed circulation, a mouth and an anus, and does not have an exoskeleton. The organism most likely belongs to the phylum: (dmc 2009)
A. Mollusca
B. Annelida
C. Arthropoda
D. Chordate

29. Which one of the following undergoes complete metamorphosis? (dmc 2007)
A. Grasshopper
B. Butterfly
C. Crab
D. Scorpion

30. Hiruden anti-clotting factor is found in: (dmc 2008)
A. Leech
B. Earthworm
C. Wuchereria
D. Nereis

31. Pheretima posthuma is the scientific name of: (lumhs 2015)
A. Planaria
B. Liver fluke
C. Tape worm
D. Ascaris
E. Earth worm

32. Example of cnidarians includes all except: (lumhs: 2009)
A. Corals
B. Jelly fish
C. Hydra
D. Sponges
E. Sea anemone

33. Which of the following is/are adaptations(s) for parasitic mode of life in Platyhelminthes? (NTS 2014)
A. Thick body covers
B. High fertility rate
C. Adhesive organ like suckers
D. Simplified digestive system
E. All of the above

34. Which group of organisms have following features? (Liyari 2014)
I. Three pairs of joined legs
II. Three-part segmented body
III. One pair of antennae
A. Arachnids
B. Myriad pods
C. Insects
D. Crustaceans

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35. Arthropods can be characterised by all of the following except. (lumhs 2007)
A. A hard exoskeleton
B. A water vascular system
C. Jointed appendages
D. Moulting

36. Which of the following endoparasites of humans replicates within human cells? (JSMU 2018)
A. Bacteria
B. Leech
C. Fungi
D. Helminths

37. Which of the following phyla of kingdom animalia does human belong to? (JSMU 2018)
A. Mollusca
B. Arthropoda
C. Enchinodermata
D. Chordate

38. Which of the following phylum and its examples is not correct? (PUMHS 2016)
A. Nemotoda .. Ascaris and hook worm
B. Annelid.. Nereis and leech
C. Echinodermata.. bath sponge
D. Cnidarian.. hydra
E. Platyhelminthess .. Planaria and liver fluke

39. Bat is: (lumhs 2011)
A. A bird
B. An insect
C. A mammal
D. A reptile
E. An amphibian

40. Annelida are: (lumhs 2011)
A. Acoelomate
B. Pseudo coelomate
C. Haemo coelomate
D. Coelomate
E. None of the above

41. A man has the following symptoms during a Filariasis:
I. Excessive inflammation of legs
II. Inflammation of arms & scrotum The disease is caused by **(lumhs 2010)**
A. A round worm (Ascaris)
B. A flat worm (Dugesia)
C. A hook worm (Ancylostoma)
D. A thread worm (Wuchereria)
E. An earth worm

42. The segmented worm belong to the phylum_____ (lumhs 2012)
A. Annelida
B. Nematoda
C. Porifera
D. Molluska
E. Arthropoda

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43. _____ are exclusively marine invertebrates. (lumhs 2012)
A. Arthropods
B. Echinoderms
C. Protozoans
D. Nematods
E. Platyhelminthes

44. Class of Annelida whose members are exclusively marine is? (lumhs 2013)
A. Polychaeta
B. Oligochaeta
C. Hirudinaria
D. None of the above

45. Pheretima Posthuma is biological name for_____ (lumhs 2013)
A. Sea anemone
B. Corals
C. Obelia
D. Jelly fish
E. Earthworm

46. Of the following, the animal not included in the same phylum as snail is the: (NTS Smmbbmu 2013)
A. Pila
B. Octopus
C. Prawn
D. Loligo
E. Pearl oyster

47. The light reaction of photosynthesis occur in the: (NTS Smmbbmu 2013)
A. Stroma
B. Thylakoid membranes
C. Mitochondria
D. Nucleus
E. Ribosomes

48. All of the following are needed for photosynthesis EXCEPT? (NTS Smmbbmu 2013)
A. Light
B. Glucose
C. Chlorophyll
D. Water
E. Carbon dioxide

49. All of the following are mammals EXCEPT the:
A. Porpoise
B. Shark
C. Whale
D. Walrus
E. Seal

50. Table shows some characteristics of four different vertebrates. Which vertebrate is a reptile? (NTS 2017)

	Fins	Legs	Scales	Hair
A.	✔	✖	✔	✖
B.	✖	✔	✔	✖
C.	✖	✔	✖	✖

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D.	✖	✔	✖	✔

Key: ✔ = feature present

✖ = feature absent

A. A
B. B
C. C
D. D

51. Platyhelminthes means: (NTS 2017)
A. Flat worms
B. Round worms
C. Segmented worms
D. None of the above

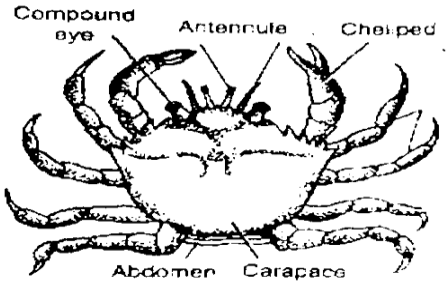
52. Hydra, Sea anemone & corals occur only as: (dmc 2010)
A. Medusa form
B. Spicules
C. Polyp form
D. Sycon form
E. None of the above

53. The proper measurements taken to avoid the diseases caused by parasitic worms include: (dmc 2010)
A. Hygienic living
B. Careful use of edibles
C. Thorough cooking of meat
D. Avoid walking bare footed
E. All of the above

54. Which statement about the phylum Echinodermata is false? (Liyari 2013)
A. The phylum includes starfish and sea urchins
B. Echinoderms usually reproduce sexually
C. The phylum includes crayfish
D. Echinoderms are heterotrophs
E. Echinoderms are invertebrates

55. Compound eyes are found in: (Liyari 2013)
A. Porifera
B. Coelenterata
C. Mollusca
D. Arthropoda
E. Annelida

56. The diagram shows an arthropod.



To which group it belongs? (pumhs 2015)

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A. Crustacea
B. Arachnids
C. Insects
D. Myriapods

ANSWERS KEY

1	D	12	B	23	C	34	C	45	E
2	B	13	E	24	D	35	B	46	C
3	C	14	E	25	B	36	A	47	-
4	A	15	C	26	B	37	D	48	-
5	B	16	C	27	B	38	D	49	B
6	B	17	D	28	B	39	C	50	B
7	C	18	A	29	B	40	D	51	A
8	B	19	E	30	A	41	D	52	C
9	B	20	C	31	E	42	A	53	E
10	D	21	D	32	D	43	B	54	C
11	C	22	D	33	E	44	D	55	D
								56	A

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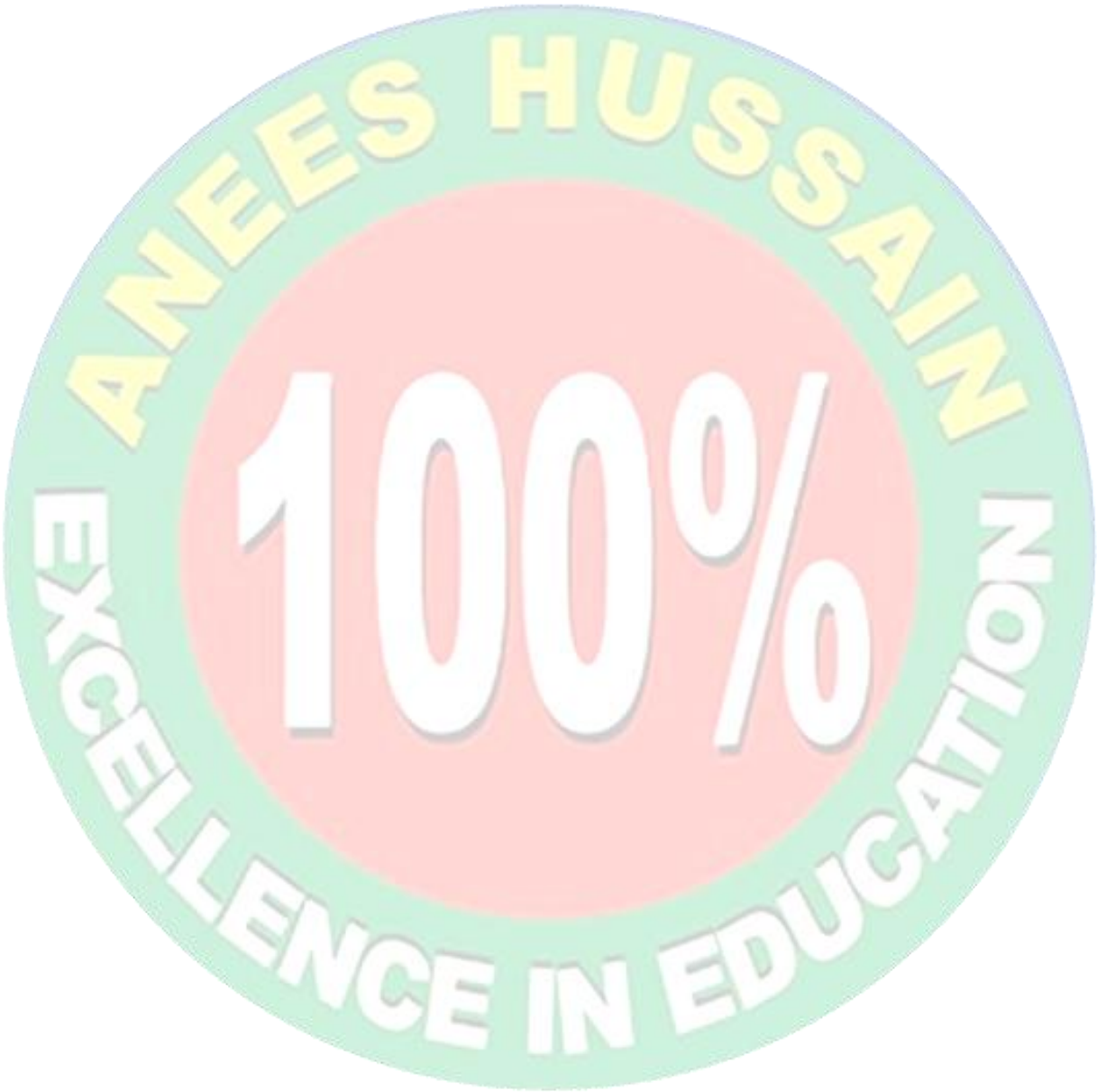

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Compiled By: **Dr. Muhammad Habib Ameen**

Bioenergetics

1. The process in green palms, in which organic food mainly carbohydrates are synthesized from water and carbon dioxide using energy absorbed from sunlight by the chlorophyll pigment is known as. (1998)
A. Transpiration
B. Dark reaction
C. Light reaction
D. Photosynthesis

2. Complete the reaction 6CO2 + 12H20→ (1998)
A. C6H12O6+6O2+6H30
B. C6H12O6+6O2+6H20
C. C5H11O3+6O2+H20
D. None

3. During glycolysis the six carbon glucose splits up a changes into two molecules of a three carbon substance called (1998)
A. Citric acid
B. Pyruvic acid
C. Tartaric acid
D. None

4. Photosynthesis is a phenomenon which converts: (1999)
A. Heat energy into chemical energy
B. Chemical energy into light energy
C. Light energy into heat energy
D. Light energy into chemical energy

5. Which of the following is not an essential requirement of photosynthesis?
A. Suitable temperature
B. Carbon dioxide
C. Water
D. Oxygen

6. The end product of glycolysis is (lumhs 2005, dmc 2011, pumhs 2015)
A. Acetyl coa
B. Pyruvic acid
C. Lactic acid
D. Citric acid

7. The electron transport chain is a group of molecules located in the (lumhs 2006 NTS)
A. Inner membrane of the mitochondrion
B. Outer membrane of the mitochondrion
C. Inter-membrane space of the mitochondrion
D. Matrix of mitochondrion

8. Enzymes associated with Krebs cycle are found in: (dmc 2006)
A. Outer layer of mitochondrion
B. Inner layer of mitochondrion
C. Inner membrane space of mitochondrion
D. Matrix of mitochondrion

9. Xanthophyll is a pigment having: (dmc 2006)
A. Yellow color

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B. Red color
C. Green color
D. Blue color

10. There are different types of chlorophyll like chlorophyll a, and: (dmc 2006)
A. B, c, d and e
C. Xanthophyll
B. B
D. Only c

11. Enzyme carbonic anhydrase in rbcs help in_____ transportation
A. Oxygen
B. Iron
C. Calcium
D. Carbon dioxide

12. Which of the following statements regarding photosynthesis is not true: (lumhs 2011)
A. The light cycle occurs only during exposure to light.
B. The dark cycle occurs only in the absence of light.
C. ATP is produced during light cycle
D. During dark cycle sugar phosphates are produced.
E. Red and blue light are optimal for photosynthesis.

13. Where do the light-dependent reactions of photosynthesis occur? (SBBMC 2016)
A. In the guard cells of the stomata
B. Stroma of chloroplast
C. In the thylakoid membrane of chloroplast
D. Cytoplasm of leaf cell

14. The dark reactions of photosynthesis are characterized by: (dmc, SMC 2016)
A. Synthesis of ATP, O2 and NADH
B. Utilization of ATP, CO2 and NADPH
C. Electron transfer from NADPH to rubp
D. Carbon dioxide transfer from rubp to glucose

15. Rubp + O2 $\xrightarrow{Rubisco}$ PGA+ _____. (dmc 2016)
A. Phasphoglycerate
B. Phasphoglycolate
C. Glycerate
D. Glycolate

16. Three word equations are shown:
P. Carbon dioxide + water → glucose oxygen
Q. Glucose + oxygen → carbon dioxide + water
R. Glucose → alcohol + carbon dioxide
What is/are the equation(s) for anaerobic respiration in yeast? (sbbmc 2015)
A. P only
B. Q only
C. R only
D. All of the above

17. In the Krebs cycle, substrate level phosphorylation accompanies the formation of: (dmc 2013)
A. Citrate
B. Alpha-ketoglutarate

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C. Succinate
D. Fumarate
E. Oxaloacetate

18. Glucose → 2pyruvic acid + 2H2O → 2ADP + 2Pi + 2ATP2 NAD + 4H → 2NADH + 2H+
These three reactions collectively constitute _____. (dmc 2012)
A. Krebs cycle
B. Calvin's cycle
C. Electron transport chain
D. Light reaction
E. Glycolysis

19. The light dependent reaction of photosynthesis occurs in: (dmc 2011)
A. Stroma of chloroplast
B. Guard cells of stomata
C. Thylakoid membrane of chloroplast
D. Cytoplasm of leaf cell
E. None of the above

20. The end product of glycolysis: (dmc 2011)
A. Citric acid
B. Pyruvic acid
C. Lactic acid
D. Acetyl coa

21. The main process that occurs in the dark reaction in photosynthesis is: (dmc 2009)
A. That water is split
B. Light energy is converted into chemical energy
C. That glucose is oxidized
D. That carbon dioxide is fixed E. None of the above

22. In paper chromatography xanthophyll will give _____ color: (dmc 2009)
A. Orange
B. Grey
C. Yellow
D. Blue-green
E. Yellow-green

23. The Calvin Cycle consist of _____ main reactions: (dmc 2009)
A. 3
B. 6
C. 9
D. 13
E. 16

24. In which of the following processes ATP is produced? (dmc 2007)
I. Photosynthesis
II. Aerobic respiration
Ill. Anaerobic respiration
A. I only
B. Il and Ill only
C. I, II and III
D. II only

25. Oxygen reacts with rubp in the absence of CO2 to form: (dmc 2007)

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A. PGA
B. PGA and phosphoglycoate
C. Oxalo acetate
D. ATP

26. Plants convert solar energy into chemical energy present in the bonds substances namely: (dmc 2008)
A. NADPH2
B. Nucleotide
C. ATP
D. RNA

27. In light reaction antennae complex _____ is not found: (2008)
A. Haemoglobin
B. Chlorophyll a
C. Chlorophyll b
D. Carotenoids

28. What happens during the light phase of photosynthesis? (NTS 2014)
A. ADP is hydrolysed and NADPH oxidized
B. ATP is synthesized by photophosphorylation and NADP reduced
C. ATP is hydrolysed and NADP oxidized
D. ATP is hydrolysed and NADP reduced
E. ATP is phosphorylated and NADP is oxidized

29. What is oxidative phosphorylation?(Liyari 2014)
A. Addition of phosphate to ADP using energy gained by transferring electrons along a chain of carriers
B. Addition of phosphate to ATP using energy gained by transferring electrons between chlorophyll molecules
C. Addition of phosphate to glucose in the first step of glycolysis
D. Removal of phosphate from ATP with the

30. Which fact about NADPH is true? It is produced: (LUMHS 2016)
A. During dark reactions and is used in dark reactions
B. During light reactions and is used in dark reactions
C. During dark reactions and is used in light reactions
D. During both reactions but is used in dark reactions only

31. Which compound capture light energy in the plants? (lumhs 2007)
A. O_2
B. CO_2
C. H_2O
D. Chlorophyll
E. None

32. The calvin - Benson cycle (lumhs 2007)
A. Does not require light directly
B. Occurs in the cytoplasm
C. Release CO_2
D. Produce ATP
E. None of the above

33. Which of the following statements about kreb's cycle is not true? (lumhs 2007)
A. The kreb's cycle occur in the matrix of the mitochondrion
B. The kreb's cycle id linked to the glycolysis by pyruvate.
C. The kreb's cycle is the single greatest direct source of ATP in the cell

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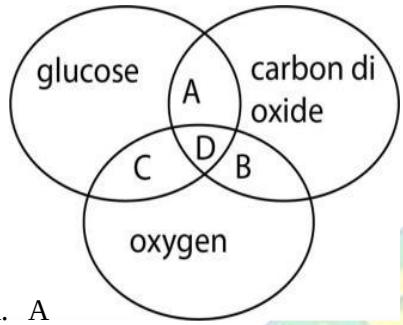
D. Citrate is an Intermediate in the kreb's cycle

E. The kreb's cycle produces nucleotides such as NADH+ H^+ and $FADH_2$

34. The diagram refers to some substances found in plant cells. Which are of the diagram represents the end products of photosynthesis? (LUMHS 2016)



A. A
B. B
C. C
D. D

35. Which part of the cell does glycolysis occur in? (JSMU 2018)
A. Mitochondrion
B. Chloroplast
C. Cytoplasm
D. Nucleus
E. Vacuole

36. The process by which one molecule of glucose splits into molecules of pyruvate is called: (JSMU 2018)
A. Glycolysis
B. Oxidation phosphorylation
C. Electron transport chain
D. Krebs cycle
E. Calvin cycle

37. Carbon dioxide joins the photosynthetic pathway at: (JSMU 2018)
A. PSI
B. PSII
C. Light reaction
D. Dark reaction

38. Plant pigments responsible for red, yellow and orange colours in many fruits and vegetables are: (LUMHS 2019)
A. Chlorophyll A
B. Chlorophyll B
C. Carotenoids
D. Cellulose

39. The complete aerobic oxidation of glucose results in the synthesis of as many as _____ molecules of ATP. (LUMHS 2019)
A. 16
B. 26
C. 36
D. 46

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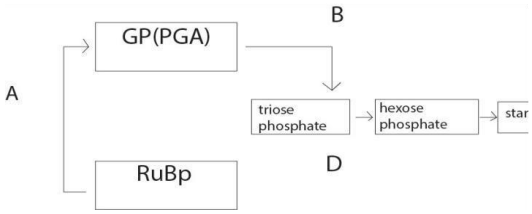
40. The diagram represents the calvin cycle at which stage is CO_2 incorporated? (kmdc 2016)



A. A
B. B
C. C
D. D

41. What happens during light phase of photosynthesis? (PUMHS 2015)
A. ADP is hydrolysed and reduced NADP is oxidised
B. ADP is phosphorylated and NADP is reduced
C. ATP is hydrolysed and NADP is reduced
D. ATP is phosphorylated and NADP is reduced

42. The end product of glycolysis is: (PUMHS 2015)
A. Acetyl coa
B. Pyruvic acid
C. Lactic acid
D. Citric acid

43. Photosynthesis pigment carotene when separated through chromatography gives _____ colour. (PUMHS 2016)
A. Grey
B. Yellow
C. Orange
D. Blue – green
E. Yellow – green

44. Carbon dioxide fixation can be enhanced by enhancing the efficiency of: (LUMHS 2019)
A. Auxins.
B. Ribulose biphosphate
C. Lactoferin
D. Agrobacterium

45. Which of the following processes has a net reaction of:
Pyruvic acid + 3H2O + 5 carries → 3CO2 + 5 carriers (2H+) (lumhs 2011)
A. Glycolysis
B. Fermentation
C. Tricarboxylic Acid Cycle
D. Electron Transport Chain
E. None of the above

46. The source of oxygen in photosynthesis is: (lumhs 2011)
A. CO2
B. Glucose
C. ATP
D. H2O
E. NADP

47. Pyurvic acid is the end product of _____ (lumhs 2012, 2017)
A. Kreb's cycle
B. Glycolysis
C. Photosynthesis
D. Anaerobic respiration
E. Electrton-transport chain

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48. Common to the kreb's cycle and glycolysis in eukaryotes is: (lumhs 2013)
A. Production of FADH2
B. Localization in mitochondria
C. Occurence in anaerobic conditions
D. Substrate-level phosphorylation
E. Oxygen utilization

49. The photosynthetic pigments can be separated by a process called: (dmc 2010)
A. Photosynthesis
B. Respiration
C. Paper chromatography
D. Hydrolysis
E. Bioenergetics

50. During regeneration phase of Calvin Benson cycle only one molecule of 3C (triose phosphate) is produced, which can:
I Re-enter the cycle
II be used for starch synthesis within the chloroplast
III exported via a phosphate translocator to cytosol for sucrose synthesis
A. I only
B. III only
C. I and III only
D. II and III only
E. I, II and III

51. Which of the following is a stage in the light independent reactions in photosynthesis (calvin cycle)? (Liyari 2013)
A. Carbon Fixation
B. Photolysis of water
C. Photophosphorylation of ADP
D. Release of oxygen
E. None of the above

52. Transport of three protons through the atpase complex are required for the production of one _____.(LUMHS 2019)
A. Sugar molecule
B. NADP molecule
C. ATP molecule
D. NADPH molecule

ANSWER KEY

1	D	12	B	23	D	34	C	45	C
2	B	13	C	24	B	35	C	46	D
3	B	14	D	25	B	36	A	47	B
4	D	15	B	26	C	37	D	48	D
5	D	16	C	27	A	38	C	49	C
6	B	17	C	28	B	39	C	50	D
7	A	18	E	29	A	40	A	51	A
8	A	19	C	30	B	41	B	52	C
9	A	20	B	31	D	42	B		
10	A	21	B	32	A	43	C		
11	A/D	22	C	33	C	44	B		

https://youtube.com/playlist?list=PLDnHG9ixGv1OkuNgFVU3Bu5inFarSs0i6

MDCAT Session 2021

Past papers since 1998-2020

Life processes in plants and animals

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Life processes in Plants and Animals

1. Which plant is insectivorous (1999)
A. Venus fly trap
B. Rose
C. Banana
D. Vinsa rosea

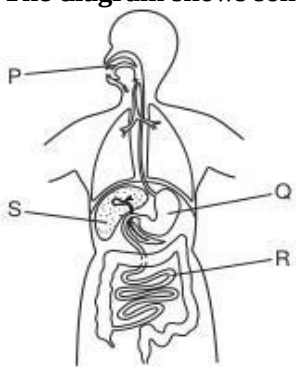
2. Sugar is stored in the liver as
A. Cellulose
B. Maltose
C. Glycogen
D. Starch

3. Herbivores depend on a diet composed of (1999)
A. Fats
B. Carbohydrates
C. Protein
D all of these

4. All of the following are parts of large intestine except (lumhs 2005)
A. Colon
B. Jejunum
C. Rectum
D. Caecum

5. Pepsin works best at a ph of _____. (Dmc, SMC 2016)
A. 1
B. 1.1
C. 1.3
D. 1.4

6. The diagram shows some of the organs of the human body.



In which organs does the digestion of carbohydrates take place? (dmc, smc 2016)
A. P and Q
B. P and R
C. Q and R
D. Q and S

7. Dietary fiber passes through several structures after leaving the stomach. In which order does the dietary fibre pass through these structures? (dmc 2015)

A. Duodenum → jejenum→ ileum→ rectum→ colon
B. Ileum→ duodenum→ colon→ jejenum→ rectum
C. Ileum→ duodenum→ jejunum→ rectum→ colon.
D. Colon→ duodenum→ ileum→ rectum→ jejunum

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E. Duodenum→ jejunum→ ileum→ colon→ rectum

8. Four words are shown below:
Facultative obligate saprophytes parasites
These words can be used in the spaces P, Q, R and S to complete the sentence below.
Among heterotrophic plants those which depend on living plants and animals for their nutritional requirements are known as ...P Parasites which depend for their nutrition entirely on other living organisms are known as ...Q... Or total parasites and those which depend for these requirements partially on other living called ...R... Or partial parasites. On the other hand, the plants which depend on dead or rotten organic remains of plants and animals are called ...S....
(dmc 2015)

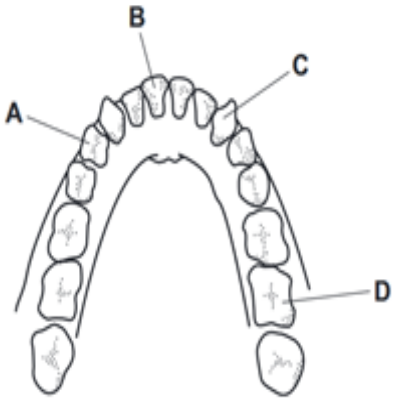
	Obligate	Parasites	Saprophyt	Facultativees
A.	P	Q	R	S
B.	Q	P	S	R
C.	R	S	Q	P
D.	S	R	Q	P

9. In an experiment, the production of hormone secretin was blocked. As a result, levels of all of the following enzymes were affected EXCEPT: (dmc 2014)
A. Trypsin
B. Pepsin
C. Chymotrypsin
D. Amylase
E. Lipase

10. Ileum is about _____ long. (dmc 2009)
A. 3.6 millimeters
B. 3.6 centimeters
C. 3.6 inches
D. 3.6 meters
E. 3.6 kilometers

11. The propulsive movement of the gastro-intestinal tract (GI tract) is: (dmc 2009)
A. Peristalsis
B. Epiglottis
C. Antiperistalsis
D. Anus
E. None of the above

12. The diagram shows teeth of lower jaws of a human. Which tooth is an premolar? (lumhs 2015, pumhs 2015)



A. A
B. B
C. C
D. D

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13. The pancreas is stimulated to secrete its digestive enzymes by: (lumhs 2015)
A. The liver
B. Bile
C. The gallbladder
D. Secretin
E. Insulin

14. A secretions that digest both carbohydrates and proteins is: (NTS 2009)
A. Ptyalin
B. Saliva
C. Pepsin
D. Gastric juice
E. Pancreatic juice

15. What role does pepsin play in digestion? (NTS 2014)
A. It kills the bacteria of stomach
B. It converts soluble milk proteins into insoluble proteins
C. It converts starch and glycogen into maltose
D. It activates the gastric glands to produce gastric juices
E. It converts proteins into short chain polypeptides

16. The bile pigment bilirubin and biliverdin are formed by the breakdown of haemoglobin of worn out R.B.C's in the _____. (lumhs 2009)
A. Spleen
B. Heart
C. Kidney
D. Lung
E. Liver

17. All of the following are functions of the liver except? (lumhs 2007)
A. Produce bile
B. Store glycogen
C. Secret insulin
D. Store vitamin

18. The enzyme found in saliva is responsible for the digestion of carbohydrates: (JSMU 2018)
A. Lysozyme
B. Amylase
C. Pepsin
D. Trypsinogen
E. Lipase

19. All of the followings are parts of large intestine except: (PUMHS 2015)
A. Colon
B. Jejunum
C. Rectum
D. Caecum

20. What role does amylase play in digestion? (PUMHS 2016)
A. It kills bacteria of stomach
B. It converts soluble milk proteins into insoluble proteins
C. It converts starch and glycogen into maltose.
D. It activate the gastric glands to produce gastric juice
E. It converts proteins into short chain polypeptides

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21. The category of organisms which are able to make their own food is called: (JSMU 2018)
A. Heterotrophs
B. Autotrophs
C. Consumers
D. Decomposers
E. Saprotrophs

22. The utilization of the absorbed food molecules within the cells to provide energy for tissue building is called (lumhs 2010)
A. Digestion
B. Assimilation
C. Egestion
D. Absorption
E. Ingestion

23. Sundew is an example of _____ (lumhs 2012)
A. Autotrophic plant
B. Parasitic plant
C. Saprophytic plant
D. Carnivorous plant
E. Extinct plant

24. After starch is ingested, in which order do these processes occur? (lumhs 2013)
A. Absorption→ assimilation→ digestion
B. Absorption → digestion→ assimilation
C. Digestion → absorption→ assimilation
D. Assimilation→ absorption→ digestion

25. Which of the following is mismatched?(lumhs 2013)
A. nervous system... Reception, conduction, integration, effect
B. Skeletal system ... Posture, protection, locomotion, blood cell production
C. Epidermis... Support, integration, exchange, filtration
D. Gastrointestinal system ... Absorption, secretion, digestion, egestion

26. Which of the following choices correctly illustrates the course that a piece of bread takes through the digestive tract? (NTS Smmbbmu 2013)
A. Mouth → trachea→ esophagus→ cardiac sphincter→ stomach→ pyloric sphincter→ small intestine→ large intestine→ rectum→ anus
B. Mouth → pharynx→ esophagus→ cardiac sphincter→ stomach→ pyloric sphincter→ small intestine→ large intestine→ rectum→ anus
C. Mouth → pharynx→ esophagus→ pyloric sphincter→ stomach→ cardiac sphincter→ small intestine→ large intestine→ rectum→ anus
D. Mouth → pharynx→ esophagus→ cardiac sphincter→ stomach→ pyloric sphincter→ small intestine→ large intestine→ anus→ rectum
E. None of the above

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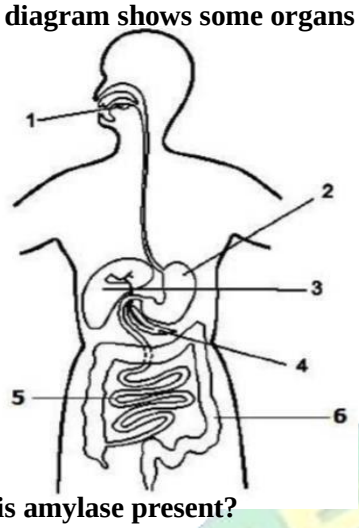
27. The diagram shows some organs of the digestive system.



Where is amylase present?

A. 1, 4 and 5

B. 1, 2 and 3

C. 2, 6 and 4

D. 3, 5 and 4

28. A part of the digestive system that is not in contact with food is the: (NTS 2017)

A. Small intestine
B. Liver
C. Stomach
D. Large intestine

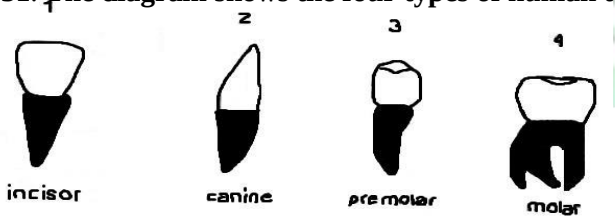
29. An example of partial root parasite is: (dmc 2010)

A. Loranthus
B. Viscum
C. Cuscuta
D. Sandal wood tree
E. Orobanche

30. The important process of holozoic nutrition includes: (dmc 2010)

A. Ingestion
B. Digestion
C. Assimilation
D. Egestion
E. All of the above

31. The diagram shows the four types of human tooth.



Which teeth are used for cutting rather than grinding food?

A. 1 and 2
B. 2 and 3
C. 3 and 4
D. 4 and 1

ANSWER KEY

1	A	9	B	17	C	25	C
2	A	10	D	18	B	26	B
3	C	11	A	19	B	27	A
4	B	12	D	20	C	28	B
5	B	13	D	21	B	29	B
6	D	14	E	22	B	30	E
7	E	15	E	23	D	31	A
8	B	16	E	24	C		

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GASEOUS EXCHANGE

1. Highly vascularised skin serve as a respiratory surface in (lumhs 2005, pumhs 2015 NTS)
A. Frog
B. Fish
C. Snake
D. Young (newly born) mammal

2. In coughing the air is forcibly blown larynx while in sneezing it is blown through: (2005 dmc)
A. Mouth
B. Nasal cavities
C. Both of these
D. None of these

3. During Inhalation

	Volume of thorax	Air pressure in lungs
A.	Decreases	Increases
B.	Decreases	Remains constant
C.	Increases	Increases
D.	Increases	Remains constant
E.	Increases	Decrease

4. Which row correctly shows the areas of the gas exchange system that contain cartilage, ciliated epithelium and goblet cells? (dmc 2016)

	Cartilage	*Ciliated*	*Goblet cells Epithelium*
A.	Bronchiole, Trachea	Bronchiole, Trachea	Bronchus, Trachea alveoli
B.	Bronchus, Trachea	Bronchiole, Bronchus	Alveoli Trachea
C.	Bronchiole, Trachea Alveoli	Bronchus, Trachea	Bronchiole, Bronchus
D.	Bronchus, Trachea	Nasal cavity, Trachea	Bronchus, Trachea

5. The adaptive feature (s) which help (s) the fish to live in water include (s):(dmc 2014)
A. A tail and an air bladder
B. Unpaired and paired fins
C. Streamed lined body
D. Gills and strong sense of smell
E. All of the above

6. Pharynx leads air into _____ through glottis. (dmc 2012)
A. Trachea
B. Bronchus
C. Alveoli
D. Nasal sac
E. Larynx

7. Haemoglobin carries more oxygen than plasma by: (dmc 2011)
A. 50 times
B. 20 times
C. 70 times

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D. 100 times
E. 200 times

8. If the level of CO2 and H+ ions increases in the body the rate of breathing will: (dmc 2008)
A. Increase
B. Decrease
C. Remains constant
D. none of them

9. During-inhalation: (NTS 2009)
A. The diaphragm flattens out and the ribs are raised
B. The diaphragm is raised and the ribs are lowered
C. The diaphragm flattens out, and the ribs are lowered
D. The diaphragm is raised and the ribs are raised
E. The diaphragm is raised and the ribs are stationary

10. Which of the following are considered to be the respiratory surfaces of the lungs? (Liyari 2014)
A. Pleural membranes
B. Pleural cavity
C. Bronchioles
D. Alveoli

11. Some characteristics of human gas exchange are listed.

1. *A difference in temperature between the air and alveoli and the blood.*
2. *A large surface area of the alveoli and the blood capillaries*
3. *The process of diffusion between the air and the blood*
4. *Elastic fibers in the walls of the alveoli*

Which of the afore mentioned factors affect the rate of oxygen uptake into the blood? (lumhs 2016)
A. 1 and 2
B. 1 and 3
C. 2 and 3
D. 3 and 4

12. Pharynx leads air into larynx through an opening called: (LUMHS 2016)
A. Glottis
B. Epiglottis
C. Trachea
D. Bronch

13. Which form is carbon dioxide mainly transported in blood (lumhs 2007)
A. As carboamino - hameoglobin
B. As carbonic acid
C. As hydrogen carbonate
D. In solution

14. Function of respiratory passages, Cilia is to keep the airways clear of: (LUMHS 2019)
A. Carbon dioxide
B. Oxygen
C. Dust
D. Carbon mono oxide

15. In the process of inspiration _____ and _____ are involved. (lumhs 2011)
A. Intercostal muscles ... Diaphram
B. Deltoid muscles ... Diaphram
C. Trapezius ... Intercostal muscles
D. Biceps ... Triceps

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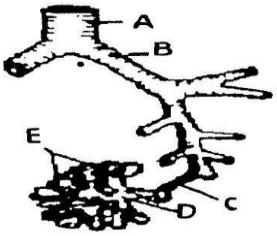
E. None of the above

16. Which is the correct sequence of passageways through which air travels during inhalation? (lumhs 2013)
A. Pharynx→ trachea→ bronchioles →bronchi→ alveoli
B. Pharynx →trachea →lungs →bronchi →alveoli
C. Larynx→ pharynx→ trachea→ bronchi →alveoli
D. Pharynx→ larynx→ trachea →bronchi →alveoli

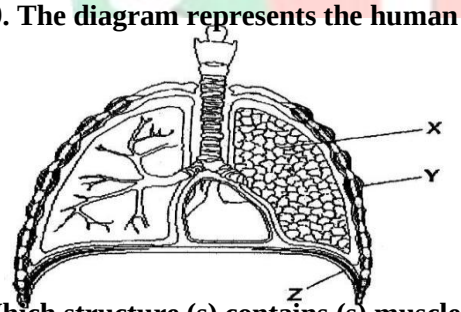
17. The absorption of oxygen from the atmosphere into the blood takes place in the: (NTS Smmbbmu 2013)
A. Pulmonary artery
B. Pulmonary vein
C. Alveoli
D. Trachea
E. Bronchi

18. In the diagram showing the bronchial tree given below, parts have been indicated by alphabets. Choose the answer in which the alphabets correctly match with the parts they indicate. (kmdc 2016)



A. A=trachea, B=bronchus, C=respiratory bronchiole, D=alveolar duct, E=alveoli.
B. A=trachea, B=bronchus, C=alveolar duct, D=respiratory bronchiole, E=atrium
C. A=bronchus, B=alveolar duct, C=respiratory bronchiole, D=trachea, E=alveoli
D. A=trachea, B=alveolar duct, C=respiratory bronchiole, D=bronchus, E=alveoli

19. The diagram represents the human respiratory system. (dmc 2014)



Which structure (s) contains (s) muscles that contract when breathing in?
A. X only
B. X and Y only
C. X and Z only
D. Y and Z only
E. X, Y and Z

20. The passageways of the respiratory system are linked by mucous secreting cells _____..(LUMHS 2019)
A. Tracheal cells
B. Goblet cells
C. Surfactant cells
D. Pleural cells

21. A _____ is an organism that makes ATP by aerobic respiration if oxygen is present, but is capable of switching to fermentation, if oxygen is absent. (LUMHS 2019)
A. Cellular anaerobe

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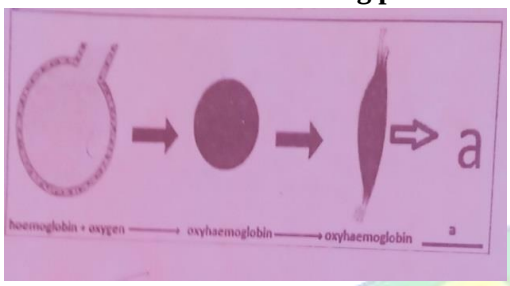
B. Respiratory anaerobe
C. Obligate anaerobe
D. Facultative anaerobe

22. What does 'a' in the following picture show? (NTS 2019)



A. Dissociation of oxyhaemoglobin
B. reassociation of hemoglobin
C. Recombination of hemoglobin
D. Breakdown of haemoglobin

ANSWER KEY

1	A	6	E	11	C	16	D	21	D
2	B	7	C	12	A	17	C	22	A
3	E	8	A	13	C	18	A		
4	D	9	A	14	C	19	D		
5	E	10	D	15	A	20	B		

TRANSPORT

1. Transpiration rates are greatest when the leaf cells are fully turgid and when the relative humidity in the atmosphere is (1998)
A. High
B. Low
C. Intermediate
D. None of these

2. Transpiration is a process of loss of water from: (1999)
A. The lower surface of leaf
B. stomata
C. The green part of the plant surfaces of the plant
D. Exposed

3. About 55% of the volume of human blood is (lumhs , dmc 2005)
A. Plasma
B. Blood proteins
C. Blood cells
D. Both b and c

4. Open circulatory system is observed in (lumhs, dmc 2005)
A. Cockroach
B. Frog
C. Fish
D. Reptiles

5. The only veins which carry oxygenated blood are (lumhs 2006 NTS)
A. Pulmonary veins
B. Interior vana cava
C. Renal veins

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D. Hepatic portal veins

6. The ability of cell to swell after absorption of H2O: (dmc 2005)
A. Inhibition
B. Deplasmolysis
C. Guttation
D. Transpiration

7. Ascent of sap is governed by the activity of the living cell in the form of pulsatory. This theory is called: (dmc 2005)
A. Physical force theories
B. Passive force theories
C. Vital force theories
D. Active force theories

8. Guttation is due to: (dmc 2006)
A. High Transpiration
B. Low Relative Humidity
C. Negative Root Pressure
D. Positive root Pressure

9. Which method of gaining immunity can be described as natural active immunity?(kmdc 2016)
A. Feeding on colostrum
B. Inhaling the chicken pox virus
C. Injection with antibodies
D. Through the placenta

10. during ventricular systole: (kmdc 2016)
A. Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary vein.
B. Oxygenated blood is pumped into the pulmonary artery and deoxygenated blood is pumped into the aorta.
C. Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary artery.
D. Oxygenated blood is pumped into the pulmonary artery and deoxygenated blood is pumped into the pulmonary vein.

11. The absorption of oxygen from the atmosphere into the blood stream takes place in the (lumhs 2011)
A. Pulmonary artery
B. Pulmonary vein
C. Alveoli
D. Trachea
E. Bronchi

12. How does sucrose move from chloroplast to phloem? (lumhs 2015)
1. Apoplast pathway
2. Symplast pathway
3. Mass flow
A. 1, 2 and 3
B. 1 and 2 only
C. 1 and 3 only
D. 2 and 3 only

13. By which process does oxygen pass out of the stem and root of a plant? (lumhs 2015)
A. Diffusion
B. Osmosis
C. Translocation

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D. Transcription

14. The following are all parts of the heart that control the heart action.

1. Purkinjie tissue
2. Sinoatrial node (SAN)
3. Atrioventricular node (AVN)

Which row for atrial contraction and ventricular contraction is correct?
(lumhs 2015)

	Atrial contraction	Ventricular contraction
A.	AVN produces wave of excitation	SAN produces wave of excitation
B.	SAN produces wave of excitation	Purkinjie tissue carries wave of excitation
C.	Purkinjie tissue carries wave of excitation	AVN produces wave of excitation
D.	SAN and AVN node produce wave of excitation	Purkinjie tissue carries wave of excitation

15. The table shows the characteristics of the blood in one blood vessel in the body.

Oxygen concentration	**Carbon dioxide concentration**	**Pressure**
High	**Low**	**High**

Which blood vessel contains blood with these characteristics?(SBBMC 2016)
A. Aorta
B. Pulmonary artery
C. Pulmonary vein
D. Vena cava

16. One type of congenital heart defect is called atrial septal defect (ASD) where the left and right atria are not completely separated. ASD usually result right atrium. This will cause increased blood pressure in the right atrium and decreased blood pressure in the left atrium.
Which of the following rows describes the other effects of ASD on blood pressure and oxygenation? (Dmc, SMC 2016)

	Blood pressure in pulmonary artery	Blood pressure in systematic aorta	% oxygenation of blood in pulmonary artery
A.	Decreased	Increased	Decreased
B.	Decreased	Increased	Increased
C.	Increased	Decreased	Decreased
D.	Increased	Decreased	Increased

17. Which of the following are found in both arteries and capillaries? (SBBMC 2015)
A. Collagen fibers

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B. Elastic fibers
C. Endothelial cells
D. Smooth muscle cells

18. About 55% 9f the blood volume is (SBBMC 2015)
A. Plasma
B. Blood proteins
C. Blood cells
D both

19. Minerals salts enter a plant largely through its: (SBBMC 2015)
A. Root hairs
B. Stomata
C. Guard cells
D. Cell wall b and c

20. The table refers to blood vessels in the Human body.

Vessels	Blood carried		Oxygenated/ deoxygenated
	From	to	
Aorta	P	All organelle except lungs	Oxygenated
Pulmonary vein	Lungs	Heart	Q
Hepatic artery	Aorta	R	Deoxygenated
Hepatic portal vein	Alimentary canal	Liver	S

What are P, Q, R and S? (dmc 2015)

	P	Q	R	S
A	Left ventricle	deoxygenated	kindey	deoxygenated
B	Left ventricle	oxygenated	liver	deoxygenated
C	Right ventricle	deoxygenated	kindey	oxygenated
D	right ventricle	oxygenated	liver	oxygenated

21. An example of passive acquired immunity is: (dmc 2015)
A. Vaccination against smallpox
B. Use of polio vaccine
C. Passing of certain antibodies to the fetus by the pregnant woman
D. Inoculation of antitoxin in case of a puncture wound
E. Both C & D

22. In the human body, blood circulating from the gut to the heart passes through the: (dmc 2013)
A. Aorta
B. Kidneys A.
C. Liver
C. Lungs
D. Spleen

23. Which valve action results from an increase in pressure in the ventricles of the heart? (dmc 2012)
A. The closing of all the heart valves
B. The closing of semi-lunar valves
C. The opening of the bicuspid valve
D. The opening of the semi-lunar valves
E. The opening of the tricuspid valve

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24. Opening and closing of stomata is controlled by which of the following taking place in the guard cells? (dmc 2007)
A. Turbidity
B. Turgidity
C. C. Flaccidity

25. Tissue fluid don't contain: (dmc 2008)
A. WBC
B. Glucose
C. Amino acid
D. Fibrinogen

26. The stomatal pore will open when the: (NTS 2009)
A: turgor of guard cells decrease
B. Turgor of guard cells increase
C. Leaf is keep in the dark
D. Amount of CO2 in the guard cells increase
E. Atmosphere becomes heavy with water vapor

27. Identify the incorrectly matched pair: (NTS 2014)
A. Cardiac output … volume of blood pumped per minute by the left ventricle
B. Atrial systole … contraction of the two atria
C. Heart murmur … defect of heart valves
D. Diastole … phase of contraction of cardiac muscle
E. Right atrium in diastolic phase …receives deoxygenated blood from vena cava

28. Why do some root cells have their root hairs? (NTS 2014)
A. For the maintenance of the temperature of the cell sap
B. To increase the surface area of the cells
C. To increase the volume of the cell sap
D. To provide a place for cell nuclei
E. To allow movement of soil particles

29. Which of the following is NOT involved in cell mediated immunity? (NTS 2014)
A. Helper T cells
B. Cytotoxic T cells
C. Suppressor T cells
D. Memory T cells
E. None of the above

30. Which statement concerning transpiration is correct?(Liyari 2014)
A. Plants transpire more rapidly in reduce light
B. Humidity and rate of transpiration are directly proportional to each other
C. Rise in temperature increase in kinetic energy of water which requires in rapid transpiration
D. None of the above

31. The blood vascular system of _____ lack hemoglobin. (lumhs 2009)
A. Humans
B. Birds
C. Fish
D. Cockroach
E. All of the above

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32. Which of the following term describe the process by which the plasma membrane moves substance inward, against a concentration gradient? (lumhs 2007)

A. Facilliated diffusion
B. Active transport
C. Osmosis
D. Simple diffusion
E. Autotrophism

33. The system responsible for fighting the pathogens is: (JSMU 2018)

A. Muscular system
B. Endocrine system
C. Nervous system
D. Immune system

34. Which of the following is NOT a part of the immune system? (JSMU 2018)

A. Phagocytes
B. Complement proteins
C. Cytokines
D. Atherosclerosis

35. A group of immune cells that mediates the cellular immune response by processing and presenting antigens for recognition by certain other cells of immune system is called _____(LUMHS 2019)

A. Natural killer cells
B. Interferon
C. Antigens-presenting cells
D. Vaccines

36. A sample of blood is taken from an unknown site in a human patient. The blood shows an oxygen Content equivalent roughly to that of venous and not the arterial circulation. Among the following, which statement is best applies to the blood that was drawn? (PUMHS 2015)

A. It was drawn from pulmonary vein and is rich in oxygen.
B. It was drawn from an alveolus and is rich in oxygen
C. It was drawn from large branches of the pulmonary artery
D. It was drawn from the superior vena cava and will enter the heart at the left ventricle.
E. It was drawn from the ventricle and will leave the heart via aorta.

37. Production of antibodies is under the control of _____(PUMHS 2016)

A. Neutrophils
B. Eosinophil
C. Lymphocytes
D. Basophils

38. During cell mediated immunity response, some T cells turn into _____ to protect the body in case of re-infection in future. (PUMHS 2016)

A. Suppressor T cells
B. Memory T cells
C. Plasma T cells
D. B cells

39. The process by which a substance takes a liquid and swells up is known as: (PUMHS 2015)

A. Guttation
B. Osmosis
C. Imbibition
D. Plasmolysis

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40. When blood is flowing through a vena cava, which main blood vessel will it flow through next? (PUMHS 2016)
A. Hepatic portal vein
B. Hepatic vein
C. Pulmonary artery
D. Pulmonary vein

41. Which of the following examples correctly match the different types of artificial immunity? (pumhs 2016)

	Active artificial	Passive natural
A	Immunity to measles after infection	Receiving antibodies to measles in colostrum
B	Immunity to measles after infection	Receiving antibodies to tetanus by injection
C	Immunity to small pox after vaccination	Receiving antibodies to measles in colostrum
D	Immunity to small pox after vaccination	Receiving antibodies to tetanus by injection

42. Opening and closing of stomata is greatly influenced by: (PUMHS 2016)
A. Concentration of oxygen
B. Concentration of hydrogen
C. Concentration of nitrogen
D. Concentration of potassium ions

43.Excess water is forced out in the form of droplets through: (lumhs 2011)
A. Stomata
B. Cuticle
C. Hydathodes
D. Lenticles
E. All of the above

44. White blood cells picking up foreign bodies from the blood stream is an example of: (lumhs 2011)
A. Pinocytosis
B. Exocytosis
C. Phagocytosis
D. Osmosis
E. None of the above

45. The possible causes of atherosclerosis are: (lumhs 2010)
A. Hypertension
B. A high serum cholesterol level
C. Family history of arterial disease
D. Physical inactivity
E. All of the above

46. The open type circulatory system is present in: (lumhs 2010)
A. Humans
B. Fishes
C. Arthropods

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D. Cow
E. None of the above

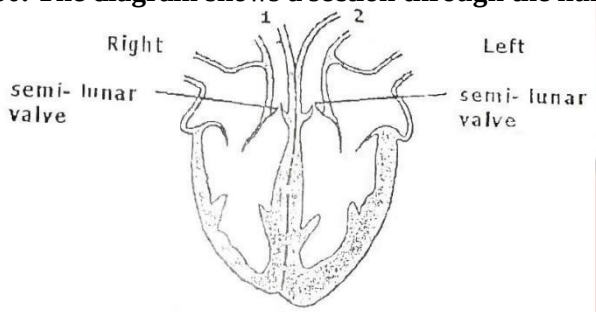
47. Pick the odd one out: (lumhs 2012)
A. Aorta
B. Arteries
C. Capillaries
D. Veins
E. Nephron

48. Which feature enables the aorta to withstand ventricular systole? (lumhs 2013)

	Semi-lunar Valves	Vessel through which blood passes to the lungs
A.	**Open**	**1**
B.	**Open**	**2**
C.	**Closed**	**1**
D.	**Closed**	**2**

49. Which cell becomes a memory cell in the immune response? (2013)
I. B-lymphocytes
II. T-lymphocytes
III. Phagocytes
A. I only
B. Ii only
C. I and II only
D. II and iii only
E. I, II and III

50. The diagram shows a section through the human heart. (lumhs 2013)



What happens as blood is being pumped out of the heart?
A. Ventricular systole
B. Ventricular diastole
C. Atrial systole
D. Atrial diastole

51. The valve present between left auricle and left ventricle is called: (NTS Smmbbmu 2013)
A. Tricuspid
B. Bicuspid
C. Auriculo-ventricular valve
D. Chordae tendinae
E. Both A & B

52. All of the following protect the body against the entrance of germs except: (NTS 2017)
A. Tears
B. Mucous membranes
C. Ciliated cells

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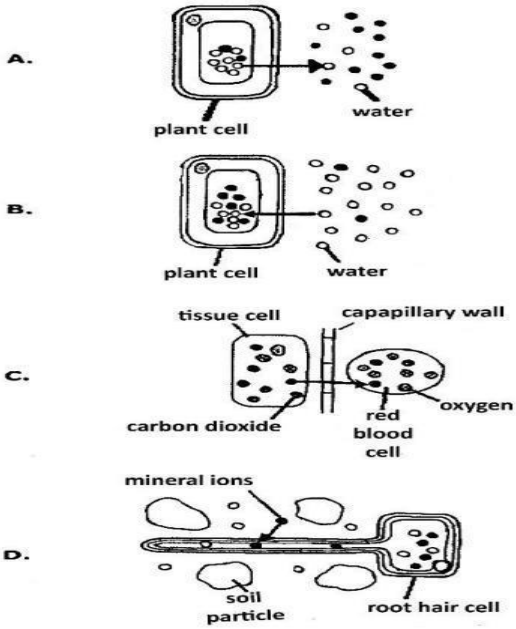
D. White blood cells
E. Red blood cells

53. Humoral immunity is carried by a special group of cells called: (NTS 2017)
A. B-cells
B. Killer cells
C. Helper cells
D. Null cell

54. Which diagram illustrates the process of active transport? (dmc 2016)

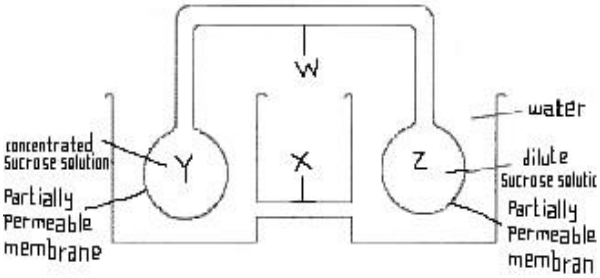


55. The diagram shows how water is lost from a leaf:
By which process is the water lost? (dmc 2014)

A. Osmosis
B. Photosynthesis
C. Translocation
D. Transpiration
E. Transcription

56. The diagram shows a model to demonstrate the mass flow hypothesis of translocation. In a plant, what are the structures w, X, Y and z and what is the direction of flow of solution along W? (dmc 2013)

	Phloem	Xylem	Roots	Leaves	Direction of flow Along W
A.	W	X	Y	Z	From Z to Y
B.	W	X	Z	Y	From Y to Z
C.	X	W	Y	Z	From Y to Z
D.	X	W	Z	Y	From Z to Y



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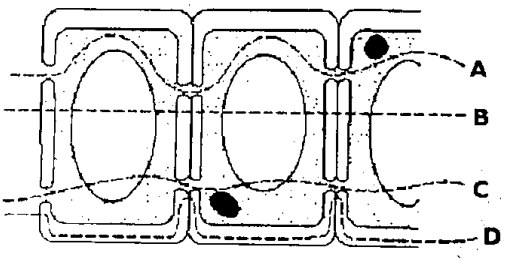
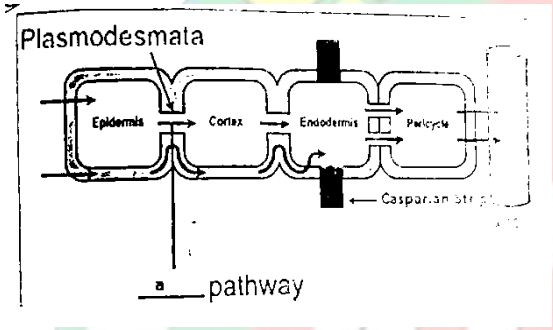
57. The diagram shows some aadjacent cells from the root of a plant. Which is Symplast pathway of water movement? (pumhs **2016)**



A. A
B. B
C. C
D. D

58. What does 'a' in the following picture show? (NTS 2019)



A. Cellular pathway
B. Synplast pathway
C. Apoplast pathway
D. water pathway

ANSWER KEY

1	A	13	A	25	D	37	C	49	C
2	B	14	D	26	B	38	B	50	A
3	B	15	A	27	D	39	C	51	B
4	A	16	A	28	B	40	C	52	E
5	A	17	C	29	C	41	D	53	A
6	A	18	A	30	C	42	D	54	D
7	A	19	A	31	D	43	C	55	D
8	B	20	B	32	B	44	C	56	B
9	B	21	E	33	D	45	E	57	D
10	C	22	C	34	D	46	C	58	B
11	C	23	D	35	C	47	E		
12	D	24	B	36	D	48	-		

SOLUTION: https://youtube.com/playlist?list=PLDnHG9ixGv1OkuNgFVU3Bu5inFarSs0i6

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Past papers since 1998-2020

Support and Movement

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Support and Movement

1. Two sides of bones in a joint attached by (2000)
A. Tendon
B. Ligament
C. Synovial cavity
D. All of these

2. Hinge joint allow the movement of the limbs _____ (2000)
A. In one direction
B. Two direction
C. Three direction
D. All direction

3. Ball and socket joint present in (2000)
A. Elbow
B. Knee
C. Shoulder
D. None

4. Synovial capsule surrounds the _____ joint.(2000)
A. Hing joint
B. Ball and socket joint
C. A and B both are
D. None of these

5. Hydrostatic skeleton can be seen in (2000)
A. Earth worm
B. Round worm
C. Flat worm
D. All of these

6. Grass hopper possess _____ type of skeleton (2000)
A. Endoskeleton
B. Exoskeleton
C. Hydrostatic skeleton
D. None of these

7. Axial and appendicular skeleton can be a part of (2000)
A. Endoskeleton
B. Exoskeleton
C. Hydrostatic skeleton
D. None of these

8. Skull, vertebral column, ribs are part of (2000)
A. Axial skeleton
B. Appendicular skeleton
C. Exo-skeleton
D. Joint skeleton

9. Pelvic, pectoral limbs are part of _____ skeleton (2000)
A. Axial
B. Appendicular
C. Hydrostatic
D. Exo

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10. Which of the following is not protected by skeleton? (2000)
A. Heart
B. Brain
C. Lungs
D. Kidney

11. Exoskeleton of arthropods is made by. (2000)
A. Caco3
B. Calcium
C. Chitin
D. Phosphate

12. The point of attachment of two bones (2000)
A. Joint
B. Synapse
C. Elbow
D. Knee

13. In vertebrates there are _____ types of muscles: (2000)
A. One
B. Two
C. Three
D. Four

14. Involuntary striated muscles are called (2000)
A. Skeleton muscles
B. Smooth muscles
C. Cardiac muscles
D. Hydrostatic muscles

15. The vertebrate skeleton is made up of (lumhs 2006)
A. Cuticle
B. Bones
C. Both A and B
D. None of the above

16. Which one of the following is voluntary muscle? (Dmc 2007)
A. Biceps
B. Cardiac
C. Stomach
D. Vessels

17. The lower two pairs of ribs are _____. (kmdc 2016)
A. True ribs
B. False ribs
C. Floating ribs
D. Articulated ribs

18. Cranium (a part of the skull) forms the brain box and consists of _____ bones: (kmdc 2016)
A. 10
B. 08
C. 05
D. 03

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19. Which of the following statements describe sliding joints? (Kmdc 2016)

I. These joints allow bone to slide over another bone to make movement in many directions
II. Vertebrae are linked by sliding joints
III. These joints don't allow the joining bones to move
IV. Bones of ankle or wrist are connected by sliding joints

A. I only
B. I & II only
C. I & III only & IV
D. I &IV only
E. I, II, III

20. Which row describes the shoulder joint of an arm? (lumhs 2015)

Shoulder joints			
	Bones	Joints type	Action
A.	Radius, ulna	Ball and socket	Flexion and extension
B.	Scapula, humerus	Ball and socket	Rotation, abduction, extension, flexion
C.	Radius, ulna	Hinge	Rotation and etension
D.	Scapula, humerous	Hinge	Flexion and etension

21. The radius and ulna join at the wrist with the _____ small carpal bones. (SBBMC 2016)

A. 4
B. 6
C. 7
D. 8

22. What is the correct sequence of the bones in the arm of a mammal, from the hand to the shoulder? (SBBMC 2016)

Hand → shoulder

A. Humerus radius scapula
B. Humerus scapula radius
C. Radius humerus scapula
D. Scapula radius humerus

23. Hinge-joint occurs in: (SBBMC 2015)

A. Elbow
B. Wrist
C. Ankle
D. Back bone

24. Identify the bones in which the connecting joints are freely moveable joints: (dmc 2015)

A. Ankle
B. Wrist

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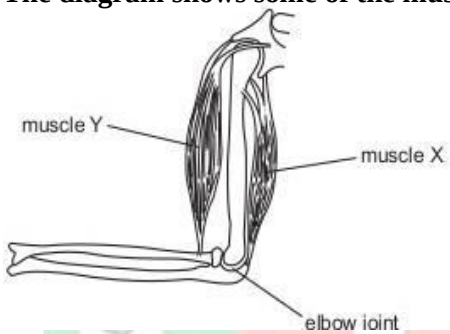
C. Vertebrae
D. Elbow
E. All of the above

25. Joints found at the vertebrae are: (dmc 2014).
A. Gliding joints
B. Sliding joints
C. Partially moveable joints
D. Did. Fixed joints
E. Pivot joints

26. When a physician elicits the knee-jerk reflex by tapping deep tendons in the knee, the normal response is for the leer leg to swing forward. When this happens: (dmc 2013)
A. Muscles in the front of the thigh are contracting and muscles in the back of the thigh are relaxing
B. Muscles in the front of the lower leg are contracting and muscles in the back of the lower leg are relaxing
C. Muscles in the back of the thigh are contracting and muscles in the front of the thigh are relaxing
D. Muscles in the back of the lower leg are contracting and muscles in the front of the lower leg are relaxing
E. None of the above

27. The diagram shows some of the muscles and bon es of the human arm. (dmc 2015, liyari 2014)



When muscle X contracts, what happens to the arm and what happens to muscle Y?

	Arm	*Muscle Y*
A.	Bends	Contracts
B.	Bends	Relaxes
C.	Straightens	Relaxes
D.	Straightens	Contracts

28. The number of muscles in a human body is about: (dmc 2012)
A. 200
B. 300
C. 400
D. 500
E. 600

29. Bones of the skull are joined by: (dmc 2011)
A. Fixed joints
B. Sliding joints
C. Pivot joints
D. Hinge joints
E. Gliding joints

30. The muscles attached to the bones are: (dmc 2010)
A. Voluntary and smooth
B. Involuntary and smooth
C. Voluntary and striated
D. Involuntary and striated

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E. Smooth and striated

31. Which of the following bones are present in the palm of hand? (dmc 2009)
A. Carpals
B. Metacarpals
C. Phalanges
D. Tarsal
E. Radius

32. In which of the following exoskeleton of glasslike silica is found: (dmc 2008)
A. Mall uses Sponges
B. Arthropoda
C. Diatoms
D. Star Fish

33. Lacteals requires which muscle to flow: (dmc 2008)
A. Unstriated and involuntary
B. Striated and voluntary
C. Striated and involuntary
D. Unstriated and voluntary

34. How many bones are present in the wrist? (lumhs 2009)
A. Four bones
B. Six bones
C. Eight bones
D. Ten bones
E. Many bones

35. Identify the correct statement: (NTS 2009)
A. Potractor and retractor muscles move the arm away
B. Abductor and adductor muscles move the arm in forward and backward direction
C. When biceps contracts elbow opens straight
D. When triceps contracts elbow bents
E. None of the above

36. The contraction of muscles depends upon (lumhs 2009)
1. Nerve impulse
2. Energy
3. Calcium ions
A. 1 only
B. 2 only
C. 1 and 3 only
D. 2 and 3 only
E. 1, 2 and 3

37. The bone at joints are held together by a tissue, the (lumhs 2007)
A. Tendons
B. Connective tissues
C. Joints
D. Ligaments

38. Which of the following statements about skeletal muscle tissue is true ? (lumhs 2007)
a. In the muscles fiber, actin is thick filament
b. The sarcoplasmic reticulum stores Ca^{+2} within a muscle cell
c. In a muscle fiber, myosin is the thin filament

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d. Contraction of a muscle fiber can occur in the absence of Ca^{+2}

39. Whenever a muscle contracts, a sarcomere can be shorten up to _____ % of its total length. (LUMHS 2019)

A. 15
B. 25
C. 35
D. 45

40. Formation of _____ with the faster break down of glucose and glycogen to compensate energy requirements in aerobic respirations. (LUMHS 2019)

A. Enzymes
B. Hormones
C. Lactic acid
D. Fat

41. The axial skeleton of an adult human includes: (PUMHS 2016)

A. The skull, pectoral girdle and ribs
B. The skull, ribs and bones of arms and legs
C. The pectoral and pelvic girdles
D. The ribs, the skull, the sternum and the vertebrae

42. The total number of bones in a human's skeleton is (lumhs 2011)

A. 207
B. 206
C. 208
D. 205
E. 203

43. Each myosin filament is surrounded by: (lumhs 2011)

A. 4 thin filaments
B. 6 thin filaments
C. 8 thin filaments
D. 12 thin filaments
E. 14 thin filaments

44. The number of tarsals and metatarsals in human feet are: (lumhs 2010)

A. 14 and 10 4
B. 16 and 10
C. 14 and 16
D. 28 and 14
E. 28 and 16

45. Which one of the following is not true? (lumhs 2012)

A. The joints of elbow and knee are hinge joints
B. The joint cavity is fitted with lubricating synovial fluid
C. Bones of the skull are connected through fixed joints
D. Hips and shoulder joints are the example of ball and socket joints
E. Cells of bones are called osteoclasts

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46. Which bones meet at the elbow joint and what kind of movement do they allow? (lumhs 2013, NTS 2017)

	BONES	MOVEMENTS
A.	Humerus and scapula	Sliding
B.	Humerus and scapula	Back and forth
C.	Ulna and humerus	Sliding
D.	Ulna and humerus	Back and forth

47. Select the correct statements: (lumhs 2013)
A. The ankle has several tarsals
B. Wrist contains nine small carpel bones
C. Fibula is the bone of thigh
D. Four metatarsals are present in the sole

48. When the knee moves back and forth as a person walks, what keeps the surfaces of the leg bones from rubbing against each other? (NTS Smmbbmu 2013)
A. The articular cartilage
B. The bone marrow
C. The periosteum
D. A sheath of smooth muscle
E. Ligaments

49. Identify the bones in which the connecting joints are immovable: (NTS Smmbbmu 2013)
A. Phalanges
B. Wrist
C. Arm
D. Leg
E. Skull bones

50. Five hip vertebrate fuse together to make: (dmc 2010)
A. Coccyx
B. Sacrum
C. Phalanges
D. Cranium
E. Femur

51. Creatine Phosphate serves as muscle's reserve of high energy phosphate by providing _____. (dmc 2010)
A. Nerve impulses
B. Acetyl chlorine
C. Atps
D. Calcium ions
E. Lactic acid

52. What is the correct sequence of the bones in the arm of a mammal, from the hand to the shoulder? (Liyari 2013)

	Hand	→ ***shoulder***
A. Humerus	radius	scapula
B. Humerus	scapula	radius
C. Radius	humerus	scapula
D. Scapula	radius	humerus

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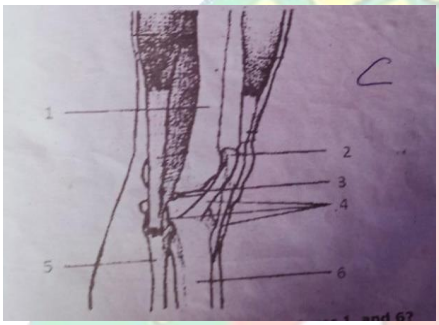
53. Pathway of energy used by muscles convert from food is: (LUMHS 2019)
A. Food-ATP-creatine phosphate-protein
B. Food-glycogen-ATP-creatine phosphate
C. Food-glycogen-creatine phosphate-ATP
D. Food-protein-creatine phosphate-ATP

54. Formation of _____ with the faster break down of glucose and glycogen to compensate energy requirements in aerobic respirations. (LUMHS 2019)
A. Enzymes
B. Hormones
C. Lactic acid
D. Fat

55. The diagram shows the joint, ligaments, bones and tendons of a human leg:



What are the correct labels for structure 1, and 6? (pumhs 2015)

	1	6
A	Fibula	Femur
B	Tendon	Patella
C	Femur	Tibia
D	Tibia	Femur

ANSWERS KEY

1	B	13	C	25	A	37	D	49	E
2	A	14	C	26	A	38	B	50	B
3	C	15	B	27	C	39	C	51	C
4	C	16	A	28	E	40	C	52	C
5	A	17	C	29	A	41	D	53	C
6	B	18	B	30	C	42	B	54	C
7	A	19	D	31	B	43	B	55	C
8	A	20	B	32	A	44	C		
9	B	21	D	33	B	45	E		
10	D	22	C	34	C	46	D		
11	C	23	A	35	A	47	D		
12	A	24	E	36	E	48	E		

https://youtube.com/playlist?list=PLDnHG9ixGv1OkuNgFVU3Bu5inFarSs0i6

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Control and coordination

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Compiled By: **Dr. Muhammad Habib Ameen**

Control and Coordination

1. The hormone which stimulate the release of an egg is known as (1998)
A. LH
B. FSH
C. LTH
D. None of these

2. The pituitary gland attached to the base is: (1999)
A. Hypothalamus
B. Adjacent to the kidney
C. Diencephalon
D. Posterior part of the Pharynx

3. Which of the following neuron received order from brain? (2000)
A. Sensory
B. Motor
C. Associated
D. Intermediate

4. Thyroxin contains_____ atoms of iodine: (2000)
A. Two
B. Three
C. Four
D. Five

5. ADH_____: (2000)
A. hormone of thyroid
B. hormone of pituitary
C. hormone of pancreas
D. None of these

6. Heart beat and breathing control ling center is: (2000)
A. Cerebrum
B. Cerebellum
C. Medulla
D. Hypothalamus

7. Hypothalamus, thalamus and cerebrum are part of: (2000)
A. Hind brain
B. Forebrain
C. Mind brain
D. None of these

8. Receptors of bitter test are located in_____ (2000
A. Tongue
B. Nose
C. Ear
D. Teeth

9. Sense of smell is transmitted to the cerebrum through: (2000)
A. Vagus nerve
B. Olfactory nerve
C. Auditory nerve

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D. Spinal nerve

10. A nerve cell that transmit s impulse from a sense organ to the nervous system by:
A. Motor neuron
B. Sensory neuron
C. Ganglion
D. None of these

11. The longer process of the cell body (2000)
A. Dendrites
B. Axon
C. Nerve
D. None of these

12. Which of the following differentiate nerve cell from general cell?
A. Lysosome
B. Mitochondria
C. Nessel substance
D. Cytoplasm

13. Synaptic vesicles present in axon terminal contain (2000)
A. Synapse
B. Nerve impulse
C. Neurotransmitter
D. Enzyme

14. When dendrites arise from the greater part of cell body of neuron is said to be (2000)
A. Unipolar
B. Bipolar
C. Multipolar
D. Myclinated

15. Thalamus and hypothalamus are part of (2000)
A. Telencephalon
B. Diencephalon
C. Mid brain
D. All of the above

16. Sleep is controlled by (2000)
A. Hypothalamus
B. Thalamus
C. Pones
D. Medulla

17. Peripheral nervous system contains (2000)
A. Cranial nerve cranial nerve
B. Spinal nerve and
C. Spinal nerve and brain
D. Spinal nerves, cranial

18. Which of the following is not a component of reflex (2000)
A. Medulla
B. Dendrites
C. Muscles
D. Effectors. The cells organs

19. Neurons that are specialized for detecting stimuli are called (lumhs 2005)
A. Nerves
B. Receptor
C. Effectors
D. All of these

20. All of the following are reflex actions except (lumhs 2005 NTS)
A. Blinking of eye
B. Knee jerk
C. Excretion of saliva upon ringing a bell without the sight of food
D. Watering of mouth on the sight of food

21. Which of the following is an effector (lumhs 2006 NTS)
A. Motor neurons
B. Sensory neurons
C. Brain
D. Gland

22. When your hand touches the hot body, which path is taken by stimulus to withdraw your hand? (Dmc 2006)
A. Sensory Neurons → Spinal Cord → Brain → Muscle.
B. Sensory Neurons → Spinal Cord → Muscle
C. Sensory Neurons → Brain → Spinal Cord →motor Neuron
D. Hot Object → Spinal Cord → Hand

23. Match the hormones listed under Column I with the roles given under Column II. Choose the answer which gives the correct combination of the alphabets of the two Columns (kmdc 2016)

Column I (Hormones)		*Column II (Roles)*
A. FSH	**P**	**Preparation of endometrium for implantation**
B. LH	**Q**	**Female secondary sexual characters**
C. Progesterone	**R**	**Contraction of uterine muscles**
D. Estrogen	**S**	**Development of corpus luteum**
E.	**T**	**Maturation of follicle**

A. A=t, B=s, C=p, D=q
B. A=r, B=t, C=s, D=q
C. A=t, B=p, C=s, D=q
D. A=q, B=s, C=p, D=r

24. The harmone progesterone: (lumhs 2011)
A. Stimulates follicle growth
B. Stimulates FSH production
C. Is solely responsible for the development of secondary sexual characteristics
D. Is produced by the anterior pituitary gland
E. Prepares the uterus for implanation

25. The pancreas is stimulated to secrete its digestive enzymes by: (lumhs 2015)
A. The liver
B. Bile
C. The gallbladder
D. Secretin
E. Insulin

26. The components of feedback mechanism are (kmdc 2016)

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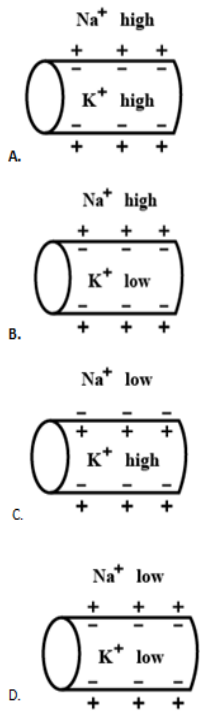
A. Receptors, Insulators, Effectors
B. Receptors, Suppressors, Effectors
C. Receptors, Control Centre, Effectors
D. Receptors, Depressors, Effectors

27. Which part of the brain regulates body temperature? (Kmdc 2016)

Column I (Hormones)		*Column II (Roles)*
A. FSH	**P**	**Preparation of endometrium for implantation**
B. LH	**Q**	**Female secondary sexual characters**
C. Progesterone	**R**	**Contraction of uterine muscles**
D. Estrogen	**S**	**Development of corpus luteum**
E.	**T**	**Maturation of follicle**

A. Cerebellum
B. Cerebrum
C. Hypothalamus
D. Medulla

28. Which diagram illustrates the distribution of sodium and potassium ions in a section of a non-myelinated axon which is at resting potential? (dmc, smc 2016)



29. Match the processes listed under Column I with the hormones given under Column II. Choose the answer which gives the correct combination of alphabets of the columns. (dmc 2016)

Column I (Processes)		Column II (Hormones)
A. Production of milk from Mammary gland	Q	Progesterone
B. Release of milk	R	Vasopressin
C. Thickening of endometrium	S	Prolactin
D. water reabsorption	T	Oxytocin

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A. A=S,B=T,C=Q,D=R
B. A=S,B=Q,C=T,D=R
C. A=T,B=S,C=Q,D=R
D. A=R,B=T,C=Q,D=S

30. Match the parts of the human brain listed under Column I with the functions given under Column II. Choose the answer

Which gives the correct combination of alphabets of the two columns. (dmc 2016)

Column I (Part of the Brain)		*Column II (Functions)*
A. Cerebral hemisphere	P	Relaying impulses
B. Thalamus	Q	Posture and balance
C. Cerebellum	R	Control of sleep and wakening
D. Medulla Oblongata	S	Reflex actions
E Hypothalamus	T	Intelligence memory

A. A=S,B=T,C=Q,D=R
B. A=T, B=P, C=Q, D=S, E=R
C. A=S,B=T,C=Q,D=R
D. A=S,B=T,C=Q,D=R, E=T

32. Which of the following is not secreted by thyroid gland? (SBBMC 2015)
A. Thyroxin
B. Triodothyronine
C. Calcitonin
D. Glucagon

33.The water content of mammalian blood is regulated by antidiuretic hormone (ADH). In which part of the nephron does regulation occur? (SBBMC 2015)
A. The ascending limb of the loop of Henle
B. The collecting duct
C. The descending limb of the loop of Henle
D. The proximal convoluted tubule

34. The diagram shows a motor nerve cell.Which part of the cell contains a transmitter substance?
A. A
B. B
C. C
D. D

35. The following sequence of events occurs at the neuromuscular junction.
Nerve impulse → release of V → end plate potential → W produced in muscle fibre → X released from sarcoplasmic reticulum → Formation of Y → muscle contraction
Which one of the following shows the correct sequence from V - Y?(dmc 2015, NTS 2017)

	V	W	X	Y
A	Acetylcholine	Action potential	Calcium ions	Actomyosin
B	Acetylcholin	Action potential	Actomyosin	Calcium e ions
C	Actomyosin	Acetylcholine	Calcium ions	Action potential
D	Calcium ions	Action potential	Acetylcholine	Actomyos in
E	Calcium ions	Actomyosin	Acetylcholine	Action potential

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36. Four events in the transmission of nerve impulses across synapses are:
I. Depolarization of the presynaptic membrane II. Propagation of postsynaptic action potential
III. Reabsorption of the transmitter substance
IV. Release of transmitter substance into the synaptic cleft
In which sequence do these events occur? FIRST → LAST (dmc 2014)

A. I III II IV
B. I IV II III
C. IV I III II
D. IV III I II

37. How many meninges cover the human brain? (dmc 2014)

A. 5
B. 4
C. 3
D. 2
E. 1

38. The autonomic nervous system controls all of the following activities except: (dmc 2014)

A. Digestion of food
B. Heart beat
C. Contraction of pupil of eye
D. Thought
E. Breathing rate

39. The gland known as the "gland of emergency" is the: (dmc 2014)

A. Pituitary
B. Adrenal
C. Thyroid
D. Parathyroid
E. Pancreas

40. What are the functions of the inter, motor and sensory neurons in a reflex response? (dmc 2013)

	INTER NEURON	MOTOR NEURON	SENSORY NEURON
A.	To connect Neurons within the Central nervous System	To conduct impulses To the Effector from the central nervous	To conduct impulses from the receptor To the central Nervous system
B.	To conduct Impulses to the Effector	To connect Neurons within the Central nervous System	To receive the Stimulus
C.	To conduct Impulses from the Central nervous System	To conduct Impulses from the Receptor to the Central nervous System	To connect neurons within the central Nervous system
D.	To conduct Impulses from The receptor to the central nervous System	To conduct Impulses from the Receptor to the Central nervous System	To conduct impulses To the effector

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41. The concentration of Na ions in the body fluids is controlled by _____ hormone. (dmc 2012)
A. ADH
B. Parathormone
C. Aldosterone
D. Estrogen
E. Thyroxin

42. The region where the impulse moves is called synapse. It consists of: (dmc 2012)
A. The axon terminals, synaptic cleft & the Dendrite of other neuron
B. Sensory neuron, motor neuron & axon
C. Myelinated & unmyelinated neurons
D. Soma & Nussle substance
E. None of the above

43. Deficient production of hormones by adrenal glands results in: (dmc 2011)
A. Cushing's syndrome
B. Addison's disease
C. Diabetes Mellitus
D. Goiter
E. Epilepsy

44. The prolactin hormone responsible for the activation of mammary glands to start producing milk is a hormone of: (dmc 2009)
A. Pituitary gland
B. Pancreas
C. Thyroid gland
D. Thymus gland
E. Adrenal gland

45. The major sign and symptom of Microcephaly is; (dmc 2009)
A. Sexual defects
B. Excessive number of toss
C. Mental retardation
D. Small skull in proportion to the normal body size
E. Split in upper lip and gap in the roof mouth

46. The automatic nervous system controls:
A. Sight
B. Thinking
C. Digestion
D. Memory

47. Which one of the following is voluntary muscle? (dmc 2007)
A. Biceps
B. Cardiac
C. Stomach
D. Vessels

48. Which is associated with long term memory? (dmc 2008)
A. Hippocampus
B. Hypothalamus
C. Brain Stem
D. Pons

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49. Pituitary gland is controlled by which of the following: (dmc 2008)
A. Adrenal Cortex
B. ACTH
C. Hypothalamus
D. Cytoxan

50. For the impulse to transfer from presynaptic to post-synaptic neuron, the substance required is: (lumhs 2015)
A. Sodium
B. Potassium
C. Protiens
D. Calcium

51. The gland that responds to external conditions of light and darkness as sensed throughout the eyes is: (lumhs 2015)
A. Thymus gland
B. Adrenal gland
C. Pineal gland
D. Thyroid gland
E. Parathyroid gland

52. _____ pairs of cranial nerves and _____ pairs of spinal nerve constitute peripheral nervous system. (NTS 2009)
A. 6 ... 12
B. 12 ... 24
C. 24 ... 12
D. 31 ... 12
E. 12 ... 31

53. A nerve cell that receives stimuli from the outside is known as a (n): (NTS 2009)
A. Motor neuron
B. Sensory neuron
C. Efferent neuron
D. Interneuron
E. Ganglion

54. When a sensory receptor receives a threshold stimulus, it will do all of the following EXCEPT: (NTS 2014)
A. Become depolarized
B. Transduce the stimulus to an action potential
C. Inhibit the spread of the action potential to sensory Neurons
D. Cause the sensory neurons to send action potentials to the central nervous system
E. None of the above

55. Destruction of all beta cells in the pancreas would cause: (NTS 2014)
A. Glucagon secretion to stop and decrease in blood glucose
B. Glucagon secretion to stop and an increase in blood glucose
C. Insulin secretion to stop and an increase in blood glucose
D. Insulin secretion to stop and a decrease in blood glucose
E. Insulin secretion to increase and decrease in blood glucose

56. during the action potential: (lumhs 2014)
A. The polarity of neurolemma first changes to -65mv and then restores to +40mv again
B. The polarity of neurolemma first changes to +40mv and then restores to -65mv again
C. The polarity of neurolemma first changes to -85mv and then restores to +30mv again
D. The polarity of neurolemma first Changes to +30mv and then restores to -85mv again

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E. The polarity of neurolemma first changes to 0mv and then restores to +40mv again

67. Adrenaline secretions increase the: (Liyari 2014)
A. Blood glucose level
B. Metabolic rate
C. Heart rate
D. Breathing rate
E. All of the above

58. Alzheimer disease is due to degeneration of _____.
A. Cartilage of bones
B. Neurons of brains
C. Blood protein factor VIII
D. Hemoglobin

59. Which of the following describes the state of no propagating (resting) mammalian axon?

	Concentration in neurolemma		Polarity of the neurolemma
	Na+	K+	
A.	High	Low	Positive
B.	High	High	Negative
C.	Low	Low	Positive
D.	Low	High	Negative

60. All of the following are true about endocrine system except (lumhs 2007)
A. It relies on chemical messenger that travels through the bloodstream
B. It is a control system that has messenger that has extremely rapid effects on the body
C. The hormones effects only certain "target" organs
D. It is involved in maintaining body homeostasis
E. Its organs secrete hormones directly into blood stream, rather than through ducts

61. The disease arises due to age is called: (JSMU 2018)
A. Alzheimer's
B. Hepatitis
C. Cholera
D. Malaria
E. Beriberi

62. As an essential element in living organisms, _____ ion is playing important role in insulin secretion, release of neurotransmitters, muscles contraction and heartbeat regulation. (LUMHS 2019)
A. Sodium
B. Potassium
C. Calcium
D. Chloride

63. Pituitary gland releases _____ hormone and _____ hormone, while ovaries produce _____ and progesterone. (LUMHS 2019)
A. Follicle stimulating and luteinizing, estrogen
B. Estrogen and follicle stimulating, luteinizing
C. Luteinizing and estrogen, follicle stimulating
D. Follicle stimulating and estrogen, luteinizing

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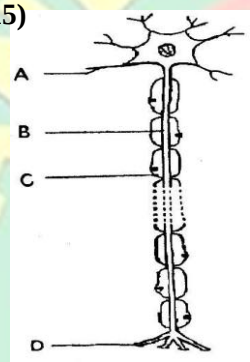
64. Synapse formed at the sites where the terminal branches of the axon of the motor neuron contact a target muscle cell are called: (LUMHS 2019)

A. Sensory end plate
B. Synapse end plate
C. Motor end plate

65. Which one is NOT true for co-ordination to animals? (LUMHS 2019)

A. Transmission by the nervous system is short lived, whereas transmission by the hormonal system is long-lasting
B. The nervous system uses electrical impulses to send signals through neurons, whereas the hormonal coordination uses chemical messengers transported into blood.
C. Responses are often permanent in the hormonal system, but temporary and reversible in the nervous system.
D. In nervous system, secretory chemicals are released in blood while in hormonal system; secretory chemicals are released in extracellular fluids.

66. The diagram shows a motor nerve cell. Which part of the cell contains a transmitter substance? (liyari (sbbmc) 2015)



A. A
B. B
C. C
D. D

67. The cells, organs or neurons that are specialized for detecting stimuli are called: (PUMHS 2015)

A. Nerves
B. Receptors
C. Effectors
D. All of the above

68. All of the followings are reflex actions except: (PUMHS 2015)

A. Blinking of eyes
B. Knee jerks
C. Excretion of saliva upon ringing a bell without the sight of food
D. Watering of mouth on the sight of food

69. A statement concerning a hormone and its production is given:
P… of cortisol results in …Q… characterized by obesity, muscle wasting, hypertension and diabetes. …R… of hormones by adrenal glands occurs due to destruction of adrenal gland. It is known as …S…, which is usually a consequence of autoimmune process. It is characterized by weakness, weight loss, and low blood sugar and reduced pressure.
Which one of the following choices shows the correct sequence and placement of terms in place of P, Q, R and S? (PUMHS 2015)

	Crushing's syndrome	Over production	Addison's disease	Deficient production

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A.	P	Q	R	S
B.	Q	P	S	R
C.	R	P	Q	S
D.	Q	R	S	P

70. The largest gland of the body is: (lumhs 2011)
A. Pancreas
B. Liver
C. Pituitary gland
D. Thyroid gland
E. Parathyroid gland

71. Which of the following statement is correct? (lumhs 2011)
A. Aldosterone is secreted by adrenal medulla
B. Parathormone is secreted by thyroid gland
C. Vasopressin is secreted by posterior pituitary gland
D. Aldosterone increases the reabsorption of potassium ions in the nephron
E. Parathormone increases the reabsorption of sodium ions in nephron

72. Blinking of eye is an example of: (lumhs 2011)
A. Taxes
B. Reflex
C. Learning
D. Insight learning E. Imprinting

73. The antidiuretic hormone is released from: (lumhs 2010)
A. Hypothalamus
B. Anterior pituitary lobe
C. Thalamus
D. The posterior pituitary lobe
E. Medulla oblongata

74. The region where the impulse moves is called synapse. It consists of: (lumhs 2010)
A. The axon terminals, synaptic cleft & the dendrite of other neuron e
B. Sensory neuron, motor neuron & axon
C. Myelinated & unmyelinated neurons
D. Soma & Nussle substance
E. None of the above

75. The phytohormones controlling the plant movement is/are: (lumhs 2010)
A. Auxins
B. Gibberellins
C. Abscisic acid
D. Cytokinin
E. All of the above

76. _____ is responsible for secondary sexual characters in male. (lumhs 2012)
A. Cortisol
B. Androgens
C. Thyroid glands
D. Glucagon
E. Aldosterone

77. Which part of the central nervous system control's the body water balance? (lumhs 2013)

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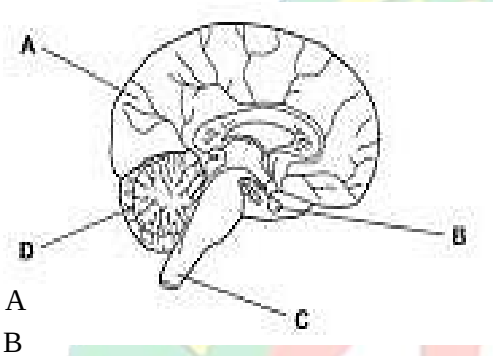
A. Cerebellum
B. Cerebrum
C. Hypothalamus
D. Medulla

78. Which of the following associations between the hormone and its role is FALSE? (lumhs 2013)
A. Estrogen-development of secondary sexual characteristics
B. Progestrone-development and maintenance of endometrial walls
C. LH-stimulate ovulation
D. FSH- formation of the ovarian follicles into the corpus luteum

79. The diagram represents a section through the brain and part of the spinal cord.
Which part of the brain control the balance?
(liyari (sbbmc 2014)



A. A
B. B
C. C
D. D

80. Adrenal gland is located: (dmc 2010)
A. At the side of the kidney
B. On the top of each kidney
C. On lower side of the liver
D. At the end of the pancreas
E. Near the gall bladder

81. _____ are the cells that separate neurons from each other and form myelin sheath. (dmc 2010)
A. Soma
B. Axon
C. Inter neurons
D. Glial cells
E. Nussle substance

82. Hyperthyroidism is always associated with: (Liyari 2013)
A. Low blood pressure
B. Severely diminished mental activity
C. Low metabolic rate
D. Low body temperature
E. Increased heart rate

83. In a reflex arc, (Liyari 2013)
A. The sensory neuron synapses directly with the motor neuron
B. Sensory and motor neuron can synapse outside of the spinal cord
C. Sensory neurons synapse in the brain
D. The motor neuron response occur without synaptic delay
E. A minimum of three neurons must participate

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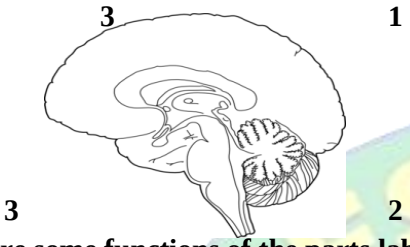
84. Which of the following associations of brain structures and function is false? (Liyari 2013)

A. Hypothalamus : appetite
B. Cerebellum : motor coordination
C. Cerebral cortex: high intellectual functions
D. Medulla : basic emotional derives

85. The diagram shows a section through the human brain.



What are some functions of the parts labeled 1, 2 and 3?

	1	2	3
A.	Heart beat and blood pressure	Forms visual Images	Controls digestion
B.	Perception of Pleasure and pain	Muscular Coordination	Heart beat and blood Pressure
C.	Muscular Coordination	Heart beat and perception	Perception of Pleasure and pain
D.	Perception of Pleasure and pain	Controls digestion	Heart beat and Blood pressure
E.	Muscular Coordination	Perception of Pleasure and pain	Heart beat and blood Pressure

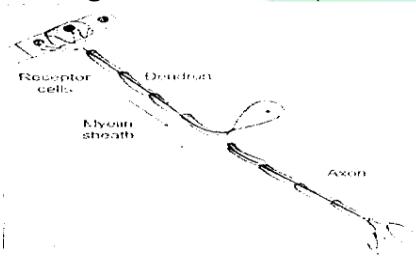
86. Which is NOT an involuntary function? (LUMHS 2019)

A. Breathing
B. Pumping of blood
C. Skeletal muscle development
D. Blinking of eye

87. All of the following are reflex action EXCEPT: (LUMHS 2019)

A. Change in the size of the pupil in response to light
B. Swallowing of bolus
C. Sudden jerky withdrawal of hand when pricked:
D. Knees jerk in a response to blow

88. What does the following structure show?(NTS 2019)



A. Motor neuron
B. sensory neuron
C. Inter neuron
D. nerve

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Reproduction

Compiled By: **Dr. Muhammad Habib Ameen**

REPRODUCTION

1. The sperm and ova are fused together to form zygote which is a result of (1998)
A. Reproduction
B. Fertilization
C. Both
D. None of these

2. In man testis are enclosed in a sac like structure known as (1998)
A. Scrotum
B. Urethra
C. Vasdeference
D. None of these

3. The follicle cells secret the female sex hormones called (1998)
A testosterone
B. Estrogen
C. Progesterone
D. None

4. In biology the types of reproduction are (1999)
A. One
B. two
C. three
D. four

5. Testes are structures that produce (lumhs 2005, SBBMC 2015)
A. Spores
B. Sperms
C. Seeds
D. Ova

6. FSH
A. Is produced in females but not in males
B. Stimulate s the seminiferous tubules to produce sperms
C. Secretion is controlled by GnRH
D. Both b and c

7. During the formation of an ovum, non-disjunction of the sex chromosomes occurred. The ovum was then fertilized by a normal, Y-bearing sperm cell. Which one of the following shows the sex chromosome complement of the resulting zygote? (dmc 2015)
A. XO
B. XY
C. XXY
D. XXXY
E. XXYY

8. All of the following are sexually transmitted diseases except: (dmc 2011)
A. Syphilis
B. Gonorrhea
C. Alzheimer's Disease
D. Genital herpes
E. AIDS

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9. The hormone which helps in the contraction of uterus at time of birth is: (dmc 2008)
A. Prolactin
B. Oxytocin
C. Gonadotropin
D. Antidiuretic hormone

10. Which one of the following is not a sexually transmitted disease? (SBBMC 2011)
A. Gonorrhea
B. Syphilis
C. Genital herpes
D. Meningitis
E. Aids

11. Spirochete bacterium Treponema pallidum causes _____. (SBBMC 2016)
A. Gonorrhea
B. Syphilis
C. Genital herpes
D. AIDS

12. The sperms produce by seminiferous tubules are stored in : (lumhs 2009)
A. Vas deferens
B. Urethera
C.Epididymis
D. Bladder
E. Sperm duct

13. All of the followings are the characteristics of the syphilis, EXCEPT (LUMHS 2016)
A. A large number of lesions are present
B. Reproductive organs are damaged along with nervous system and skeletal system
C. It is caused by the bacterium Neisseria gronorrhoeae
D. Prolonged antibiotic treatment is the possible remedy

14. Referring to sexual production, humans are: (JSMU 2018)
A. .hermaphrodites
B. Viviparous
C. Oviparous
D. Self-fertilized

15. In sexual reproduction, sex cells contains _____ number of chromosomes when compared to other cells of body. (LUMHS 2019)
A. Same
B. Half
C. Double
D. All of the above

16. Which of the followingg is/are function(s) of semen?
I. It provides the sperms with food
II. It provides the sperms with a medium to swim in
III. It neutralizes the acidity of urethra caused by the passage of Urine
(lumhs 2010)
A. I only
B. III only
C. I and II only
D. II and III only
E. I, II and III

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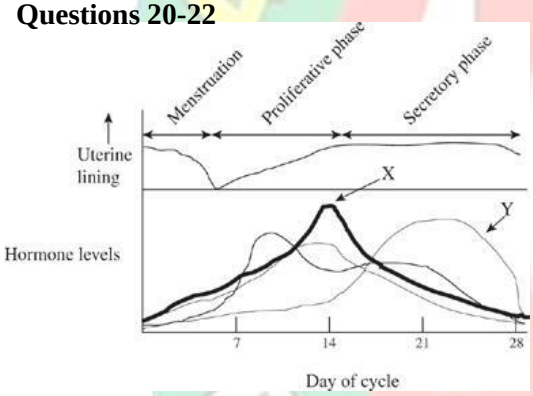
17. The strong contractions of uterus muscles are controlled by a hormone called _____ (lumhs 2012)
A. Cortisol
B. Progesterone
C. Insulin
D. Testosterone
E. Oxytocin

18**. On which date is a woman most likely to ovulate if the first day of menstrual loss was 1st February. (lumhs 2013)**
A. 5 february
B. 14 february
C. 24 february
D. 28 february
E. 1 march

19. Which of the following incorrectly pairs the term with its definition? NTS Smmbbmu 2013)
A. Scrotum-location of the testes
B. Epididymis-site of sperm maturation
C. Vas deferens-tube connecting the epididymis to the prostate
D Penis- sperm storage
E. Urethra-tube for urinary & reproductive discharge

Questions 20-22



20 Tests to determine when ovulation has occurred. This hormone is
A. Estrogen
B. Progesterone
C. FSH
D. LH
E. Testosterone

21. Based on the peak levels of hormone X, on what day of the cycle is ovulation most likely to occur?
A. Day 21
B. Day 14
C. Day 12
D. Day 25
E. Day 28

22. The hormone labeled Y in the diagram is:
A. Progesterone, secreted by the corpus luteum after ovulation has occurred
B. Progesterone, secreted by the ovary after ovulation has occurred
C. Estrogen, secreted by the corpus luteum after ovulation has occurred
D. Estrogen, secreted by the ovary after ovulation has occurred
E. Estrogen, secreted by the follicle before ovulation occurs

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23. Fertilized ovum in human is planted in: (dmc 2010)

A. Oviduct
B. Uterus
C. Placenta
D. Umbilical cord
E. Vagina

24. The contraction of uterus during labour and release of milk from the mammary glands during breast feeding is stimulated by: (dmc 2010)

A. Prolactin
B. Oxytocin
C. Antidiuretic hormone
D. Somatotropin hormone
E. FSH (follicle stimulated hormone)

25. In gametogenesis, which resultant product is non-functional? (LUMHS 2019)

A. Spermatogonia
B. Oogonia
C. Polar body
D. Ovum

ANSWER KEY

1	B	6	D	11	B	16	E	21	B
2	A	7	C	12	C	17	E	22	A
3	C	8	C	13	A	18	B	23	B
4	B	9	B	14	B	19	D	24	B
5	B	10	D	15	B	20	B	25	C

https://youtube.com/playlist?list=PLDnHG9ixGv1OkuNgFVU3Bu5inFarSs0i6

MDCAT Session 2021

Past papers since 1998-2020

Variation and Genetics

MDCAT Session 2021

Compiled By: **Dr. Muhammad Habib Ameen**

Variation and genetics

1. Color blindness and hemophilia in man are controlled by (1998)
A. Dominant genes
B. Recessive genes
C. Both of these
D. None of these

2. The condition in which blood fails to clot after injury is known as (1998)
A. Colour blindness
B. Hemophilia
C. Anemia
D. Leukemia

3. Which one of the following is called father of genetics
A. Lamarck
B. Darwin
C. Mendel
D. Hooke

4. The blood grouping in human being is due to (1999)
A. Double alleles
B. Mono alleles
C. Multiple alleles
D. None of these

5. Monohybrid cross is a cross between two individuals that differ in
A. A single trait
B. Many traits
C. Both of these
D. None of these

6. Mandel concluded that each organism has two hereditary factor for each trait, now called (lumhs 2005)
A. Chromatids
B. Allele
C. Chromosomes
D. None

7. A man who is color blind marries a normal woman, and together they have a daughter who is not color blind. If the daughter marries a man with normal vision, what is the probability of their first born child being a son who is color blind ? (lumhs 2006)
A. 0%
B. 25 %
C. 50%
D. 75%
E. 100%

8. In man tallness is _____ to shortness.
A. Dominant
B. Recessive
C. Incompletely dominant
D. Homozygous

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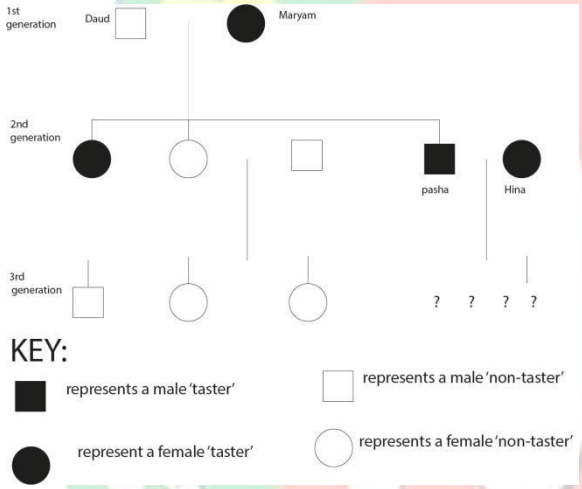
9. In a population 36% people have blue eyes"; allele frequency for blue is (dmc 2006)

A. 36%
B. 40%
C. 60%
D. 64%

10. A man with blood group 'a' marries a woman with blood group 'b', child can have (dmc 2006)

A. A and b
B. B and ab
C. A and ab
D. A, b, ab and o

11. The family tree shows the inheritance of the ability to taste a certain substance. The allele for the ability to taste this substance is dominant to the allele for the inability to taste it. (dmc 2016)



What percentage of children of pasha and hina would be 'non tasters'?

A. 25 %
B. 50%
C. 75%
D. 100%

12. Which statement about dominant and recessive alleles is not correct?(sbbmc 2016)

A. A dominant characteristic is seen in the phenotype of a heterozygote
B. A homozygous genotype may be either Dominant or recessive
C. Recessive phenotypes always have two recessive alleles.
D. The phenotype of a homozygote is always dominant

13. An Rh negative woman is married to an Rh positive man whose father was also Rh negative. What are the chances that theirchild will be affected? (sbbmc 2016)

A. 25%
B. 50%
C. 75%
D. 100%

14. Inheritance of acquired characteristics is based on: (dmc, smc 2016)

A. Genetic role in reproduction

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B. Use and disuse of organ
C. Survival of the fittest
D. Mutations

15. Two animals are mated. One is homozygous dominant for one character and homozygous recessive for another. The other animal is heterozygous for both characters. How many phenotypes are expected in the offspring of this cross? (dmc 2016)
A. 1
B. 2
C. 3
D. 4

16. Red-green colour-blindness is a sex-linked recessive trait. A colour-blind man marries a woman with normal vision, whose father is colour-blind. If they have a daughter, what is the probability that she will be colour-blind? (sbbmc 2015)
A. 0
B. 0.25
C. 0.50
D. 0.75

17. Mendel concluded that each organism has two hereditary factors for each trait, now called:
A. Chromatids
B. Alleles
C. Chromosomes
D. None of the above

18. Flower color is controlled by a single pair of alleles. The allele for red flowers is dominant to the allele for white flowers.
A plant homozygous for red flowers is crossed with a plant homozygous for white flowers. All the resulting plants have red flowers (f1 generation).
When the f1 generation is crossed with each other, 18 plants are obtained. 12 plants have red Flowers and 6 have white flowers (f2 generation)
What ratio is expected in the f2 generation and what ratio has been obtained? (dmc 2014)

	Expected ratio red to white	Obtained ratio red to white
A.	1 : 1	2 : 1
B.	1 : 1	3 : 1
C.	3 : 1	2 : 1
D.	3 : 1	3 : 1

19. In birds the male is the homogametic sex. A male bird showing the recessive trait was mated with a female showing the dominant trait of a characteristic governed by a pair of alleles which are sex linked. What is the probability that the male offspring will show the dominant trait? (dmc 2014)
A. 0
B. 0.25
C. 0.50
D. 0.75
E. 1.00

20. A pure-breeding plant with the dominant phenotype of character p and the recessive phenotype of character q was crossed with another pure-breeding plant with the. Recessive phenotype of character p and the dominant phenotype of q. The offspring of this cross were crossed with a double homozygous recessive for p and q and the following results obtained:
22 were phenotypically dominant for p and recessive for q.
5 were phenotypically dominant for both p and q.

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4 were phenotypically recessive for both p and q. 24 were phenotypically recessive for p and dominant for q. Which one of the following types of inheritance is illustrated by these results? (dmc 2013)

A. Gene linkage of p and q
B. Independent segregation of p and q
C. Mendelian dihybrid inheritance
D. Multiple allele

21. In cats, the genes controlling coat-colour are co-dominant (incompletely dominant) and are carried on the x chromosomes. When a black female was mated with a ginger male the resulting litter consisted of black male and tortoise-shell female kittens. What phenotypic ratio would be expected in the f2 generation? (dmc 2012)

A. 1 black male: 1 ginger male: 2 tortoise-shell females
B. 1 black male: 1 ginger male: 1 tortoise-shell female: 1 black female
C. 2 black males: 1 tortoise-shell female: 1 ginger female
D. 1 black male: 1 tortoise-shell female: 1 ginger female: 1 black female
E. 2 black males: 1 tortoise-shell female: 1 black female

22. Erythroblastosis foetalis occurs when: (dmc 2011)

A. Mother is rh positive and baby is rh negative
B. Mother is rh negative and baby is rh positive
C. Both mother and baby are rh negative
D. Both mother and baby are rh positive
E. All of the above statements are true

23. In pea plants, the allele for round seeds (r) is dominant to the allele for wrinkled seeds (r) and the allele for yellow seeds (y) is dominant to the allele for green seeds (y). A doubly heterozygous, round, yellow-seeded plant is crossed with a green, wrinkled- seeded plant. What percentage of the f1 generation are recombinants? (dmc 2016)

A. A.0 %
B. B.25 %
C. C.50 %
D. D.75 %
E. E.100 %

24. The law of dominance is illustrate in the garden pea by: (dmc 2009)

A. Homozygous tall x heterozygous tall
B. Heterozygous tall x heterozygous tall
C. Homozygous tall x homozygous tall
D. Pure short x pure short
E. Homozygous tall x pure short

Questions 28-30

Hemophilia is a disorder in which blood fails to clot. Saad, a male hemophiliac, marries sara, a normal woman and together they have four children, two boys (ahmed and ali) and two girls (alia and ayesha). None of the children display the symptoms of hemophilia. Ahmed, ali, alia, and ayesha all marry normal individual and have children. None of ahmed's or ali's children, male or female, display symptoms of hemophilia, but the sons of alia and ayesha display symptoms of hemophilia while the daughters of alia and ayesha do not 60.

25. Which of the following best explains the reasons that ahmed, all, alia, and ayesha do not display symptoms of hemophilia, even though their father, saad, is a hemophiliac? (dmc 2009)

A. Hemophilia is a y-linked disorder, and saad can only pass on his y chromosome.
B. Hemophilia is an x-inked disorder, and even though alia and ayesha received a hemophiliac x chromosome from saad, sara gave them a normal x chromosome.
C. Hemophilia is a y-linked disorder, and therefore cannot be displayed in females.
D. Hemophilia is a y-linked disorder, and ahmed and all must have received and x-chromosome from saad.

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E. Hemophilia is an x-inked disorder, and even though ahmed and ali received a hemophiliac x chromosome from saad, gave them a normal x chromosome.

26. If one of ali's daughters marries a normal man, what is the probability that one of their children will display symptoms of hemophilia?
A. 0%
B. 25%
C. 50%
D. 5%
E. 100%

27. Which of the following individuals are heterozygous for hemophilia?
A. Saad, ahmed and ali
B. Ahmed, ali alia, and ayesha
C. Saad and sara
D. Alia and ayesha E. Ahmed and ali

28. O group is universal donor because (dmc 2007)
A. It has both a & b antigens
B. No antigens
C. A antigen
D. B antigen

29. Skin color, height and intelligence vary in different people due to: (dmc 2007)
A. Pleiotropy
B. Epistasis
C. Polygenic inheritance
D. Crossing over

30. When phenotypically tall plant is crossed with pure dwarf plant what cross is this: (dmc 2008)
A. Test
B. Monohybrid
C. Dihybrid
D. Multiple cross

31. When hybrids are crossed, the genotype ratio of the offspring is: (nts 2009)
A. 1:1
B. 3:1
C. 1:2:1
D. 4:1
E. 3:2:1

32. A woman with normal color vision, whose father was red green color blind, married a red green color blind man. What is the probability of her first born child being red-green color blind? (nts 2014)
A. 1.0
B. 0.75
C. 0.66
D. 0.50
E. 0.25

33. The four children of two parents each have a different blood group of abo series. What were the blood group of the parents: (liyari 2014)
A. A and a
B. A and b
C. A and o

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D. B and o

34. Which of the following disease occurs due to autosomal recessive allele?
A. Cystic fibrosis
B. Huntington
C. Haemophilia

35. In man, a, b, ab , o blood groupings are controlled by a system of multiple alleles. A man of blood group a marries a woman of blood group b and they have one child. Which one of the following statements about the child's blood is correct? (lumhs 2016)
A. It could only be group A
B. It could only be group A or group B
C. It could only be group AB
D. It could not be group O
E. It could be any of the groups a, B, AB, O
F. HIV

36. Albinos have a genotype of aa, while all other members of population are either aa or aa. The offspring of a cross between a heterozygous male and albino female would be (lumhs 2007)
A. 100% albino
B. 100% normal
C. 50% normal, 50% albino

37. A man who has type ab blood could not father a child with type _____ blood, because he would pass on either the _____ or the b allele to all of his offspring. (lumhs 2019)
A. A, o
B. O, a
C. B, o
D. B, a

38. When a colour male marries a normal female, what will be the chances of colour-blindness in his grandsons, if his daughter marries to a normal male? (lumhs 2019)
A. 10%
B. 25%
C. 50%
D. 100%

39. A cross between a black cat and a tan cat produces a tabby pattern (black and tan fur together). What percent of kittens would have tan fur if a tabby cat is crossed with a black cat? (lumhs 2019)
A. 100%
B. 50%
C. 25%
D. 0%

40. The following results of a cross between twoindividuals shown in the picture is: (lumhs 2019)

	b	b
B	Bb	Bb
b	bb	bb

A. One that is homozygous dominant and other has a dominant phenotype, but has a mother with recessive phenotype
B. One that is homozygous recessive and other has a dominant phenotype, but has a mother with recessive phenotype
C. One that is homozygous recessive and other has a dominant phenotype, but has a brother with recessive phenotype.
D. One that is homozygous recessive and other has a dominant phenotype, but has a father with recessive phenotype

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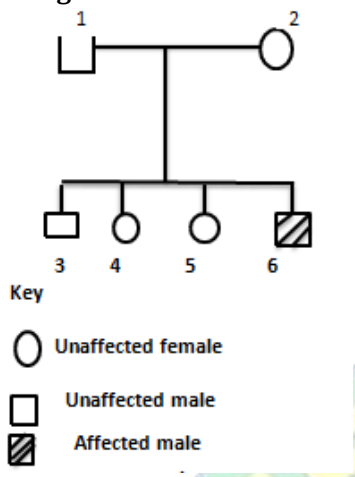
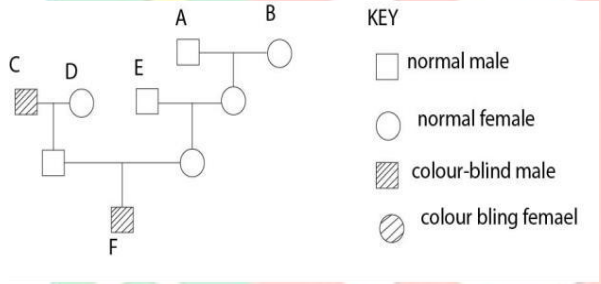
41. The diagram shows the inheritance of haemophilia in a family.



If daughter 4 married a normal male, what is the probability that their first child would suffer from haemophilia? Kmdc 2016)

A. 0
B. 0.125
C. 0.25
D. 0.5

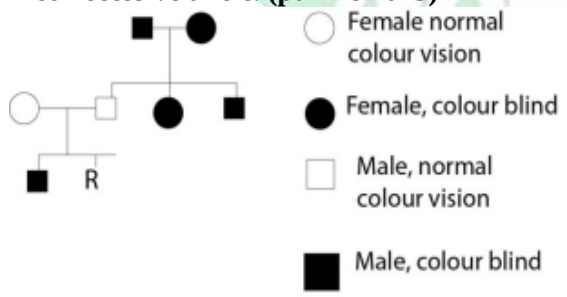
42. The diagram shows the pedigree of a family carrying the sex-linked allele for colour blindness. (lumhs 2015)



From labelled member of his family did f inherit his disorder?

A. A
B. B
C. C
D. D
E. E

43. The diagram shows the inheritance of red-green colour blindness in a family. This condition is caused by a sex-linked recessive allele. (pumhs 2015)



What is the probability that individual r will be a carrier of color blindness? (pumhs 2015)

A. 0
B. 0.125
C. 0.25
D. 0.5

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44. A man who is normal for colour vision marries a normal heterozygous woman. What is the chance of their son being colour blind (lumhs 2011)
A. 0,%
B. 25%
C. 50%
D. 75%
E. 100 %

45. Mendel perform his famous experiments of heredity on: (lumhs 2011)
A. Maize
B. Bean plants
C. Garden pea
D. Mirabilis plant
E. Wheat

46. A man who is normal for color vision marries a normal heterozygous woman. What is the chance of their son being color blind? (lumhs 2011)
A. 0%
B. 25%
C. 50%
D. 75%
E. 100%

47. Which one of the following genotype cannot occur amongst the offspring from a mating between a person of blood group a and a person of blood group b? (lumhs 2012)
A. Aa
B. Ab
C. A
D. B
E. O

48. In fruit flies, the gene for wing type is located on an autosomal chromosome. The allele for wild-type wings is dominant over the allele for vestigial wings, if a homozygous dominant male fly is crossed with a female with vestigial wings, what percentage of their female progeny is expected to have wild-type wings? (lumhs 2012)
A. 0%
B. 25%
C. 50%
D. 75%
E. 100%

49. According to normal genetic prediction a color-blind girl (xc xc) must have had a father whose genetic makeup was: (lumhs 2013)
A. Xc xc
B. Xn yc
C. Xc yc
D. X y
E. Xc y

50. In cats, the gene controlling coat-color are co dominant (incompletely dominant) and are carried on the x chromosome, when a black female was mated with ginger male the resulting litter consisted of black male and tortoise-shell female kittens. (nts smmbbmu 2013)
What phenotype ratio would be expected in the f2 generation?
A. 1 black male: 1 ginger male: 2 tortoise -shell females
B. 1 black male: 1 ginger male: 1 tortoise-shell male: 1 ginger female: 1 black female

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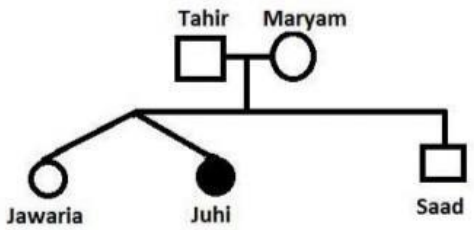
C. 2 black males: 1 tortoise-shell female: 1 ginger female
D. 1 black male: 1 tortoise-shell male: 1 ginger female: 1 black female
E. 2 black males: 1 tortoise-shell female:1 black female

51. What are the phenotypes of the parents of a color blind son and non-carrier daughter with normal color vision? (nts smmbbmu 2013)

Father	***Mother***
A. Carrier	normal
B. Color-blind	carrier
C. Color-blind	color-blind
D. Normal	carrier
E. Normal	color-blind

52. In the pedigree of a family shown above, brown eyes are indicated as ○ and blue eyes as ● Jawaria and Juhi are twins. From this chart, it can be determined that:



A. Tahir and Mary are homozygous for brown eyes
B. Jawaria and Juhi are identical twins
C. Juhi is heterozygous for blue eyes
D. Juhi is homozygous for blue eyes
E. Jawaria and saad are homozygous for brown A. Eyes

53. Which of the following RNA sequences would be transcribed from the dna sequence ATGCCTAGGAC? (NTS 2017)
A. TACGGATCCTG
B. UAGCGAUCCUG
C. AUGCCUAGGAC
D. UACGGAUCCUG
E. GACUUCGAAGU

54. If the new born babies get mixed up in a hospital, how could you determine their parentage from the information given below? (dmc 2010)

Baby I Type O
Baby II Type B
Mrs. Ali Type A Mr. Ali Type AB
Mrs. Ahmad Type A Mr. Ahmad Type A

A. Baby I is the child of Mr. And Mrs. Ahmad
B. Baby I is the child of Mr. And Mrs. Ali
C. Baby II is the child of Mr. And Mrs. Ahmad
D. Both Baby I and Baby II are the children of Mr. And Mrs. Ali
E. Inadequate Data

55. Hemophilia is caused by a sex -linked, recessive allele. Two parents have a Hemophiliac son, a normal son a Hemophiliac daughter.
What are the most likely genotype of the parents? (dmc 2010)
Mother Father
A. XH Xh Xh Y

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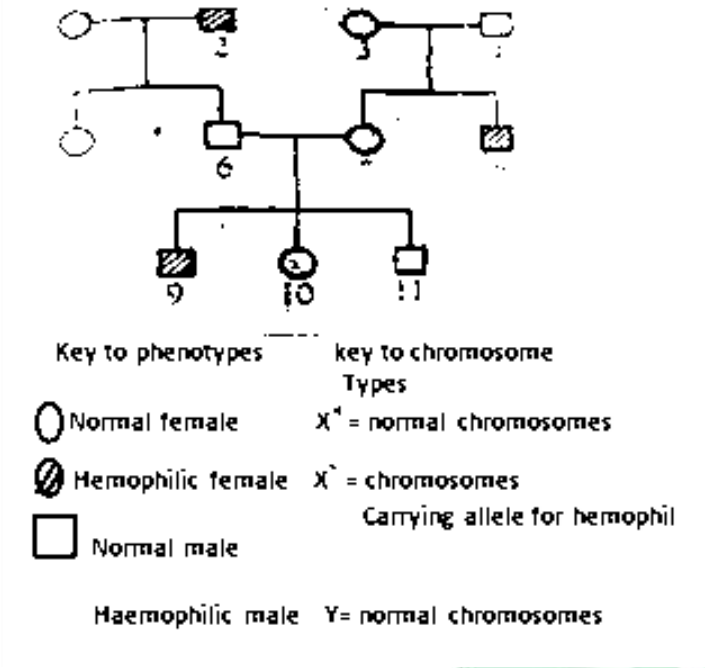
B. XH Yh XH Y
C. Xh Xh XH Y
D. XH YH Xh Y
E. None of the above

56. In a laboratory while working on a new species of fish, it is found that the fish has two varieties, red and brown. It was determined by another group of scientists in another laboratory that brown is a dominant colour in species. If we have brown fish with us in laboratory, how can we determine whether they are homozygous or heterozygous for the trait? (LUMHS 2019)

A. Breed this fish with a red fish and check f1 generation.
B. Breed this fish with a red fish and check f2 generation
C. Breed this fish with a brown fish and check f1 generation
D. Breed this fish with a brown fish and check f2 generation

57. Hemophilia is a sex linked _____ trait. (LUMHS 2019)
A. Dominant
B. Co dominant
C. Pleitropic
D. Recessive

58. The diagram shows the inheritance of haemophilia in a family. (dmc 2015)



What is the genotype of person 7?
A. XH XH
B. XH Y
C. XH Xh
D. Xh Yh
E. Xh Y

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Past papers since 1998-2020

Evolution

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Compiled By: **Dr. Muhammad Habib Ameen**

Evolution

1. Who believed that living organisms gradually evolved into new organisms as nature tended to change from simple, imperfect form to complex and perfect form (1998)
A. Aristotle
B. Darwin
C. Lamarck
D. None

2. According to the theory of evolution, the birds have evolved from
A. Reptiles
B. Amphibians
C. Fish
D. Tetrapods

3. According to lamarck, evolution occurred as the result of :
A. Natural selection
B. The theory of the recapitulation
C. Over reproduction
D. acquired character

4. Which of the following structure of fish is homologous with the lung of frog (lumhs 2006)
A. Gills
B. Swim bladder
C. Stomach
D. Book lung

5. All of the following statement s about evolution are true except (lumhs 2006 nts)
A. Living things have changed
B. Evolution has taken place over many millions of years
C. Evolution is still going on
D. Evolution has ceased

6. An inter-breeding population of finchesbecame separated geographically, forming two isolated groups. Each group then became subject to different selective pressures. One group was then introduced into the habitat of the other.which one of the following would determine whether they now formed two distinct species? (kmdc 2016)
A. They had been separated for more than three million years.
B. They failed to produce fertile f, hybrids.
C. They showed marked differences in the shape of their beaks.
D. Their plumage had become markedly different.
E. Several genes now possessed different base sequences

7. Lamarck theory is based on all of the following points except: (lumhs 2015, dmc 2011)
A. Effects of environment
B. Use and disuse of organ
C. Natural selection
D. Inheritance of acquired character

8. In the hardy-weinberg principle, $p^2 + 2pq + q^2 = 1$, q^2 represents the frequency of the: (sbbmc 2016, dmc 2009)
A. Homozygous dominant
B. Heterozygous dominant
C. Heterozygous recessive

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D. Homozygous recessive
E. Blended genes

9. The following observations refer to evolution:

I. Inherited variations which are 'favoured' in particular environment are passed on.
Ii. There is a struggle for existence.
Iii. In time, 'favoured' inherited variations may accumulate causing gradual changes in the organism.
Iv. Although populations tend to overproduce, they remain more or less constant in numbers from generation to generation.

In what sequence should the statements be placed to support. Darwin's theory of evolution? (dmc 2015, 2014)
A. I, ii, iii, iv
B. Ii, i, iii, iv
C. Iii, i, iv, ii
D. Iv, ii, i, iii

10. At what point are two populations descending from the same ancestral stock considered separate species? (dmc 2014)
A. When they can no longer produce viable, fertile offspring
B. When they look significantly different from each other
C. When they can interbreed successfully and produce offspring
D. When their habitats are separated by a significantly large distance so that they cannot meet
E. Both b & c

11. An inter-breeding population of finches became separated geographically, forming two isolated groups. Each group then became subject to different selective pressures. One group was then introduced into the habitat of the other.which one of the following would determine whether they now formed two distinct species? (kmdc 2016)
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D. Their plumage had become markedly different.
E. Several genes now possessed different base sequences.

12. Identify the incorrect statement about charles darwin's theory: (dmc 2009)
A. The individuals of species have variations among them
B. There is always a tendency of over reproduction in a species
C. Vast gradual changes result in the origin of a new species
D. Favorable variations survive and unfavorable will be exterminated
E. Intra specific competition occurs between different species and inter-specific competition occurs among the individuals in a species

13. The theory of inheritance of acquired characters was proposed by: (dmc 2008)
A. Devaries
B. Darwin
C. Lamarck
D. Weisman

14. Which of the following negates lamarck's theory? (lumhs 2009)
A. Elongated fore-limbks and neck of giraffe
B. Custom of circumcision
C. Evolution of snake
D. Foot of modern horse
E. Webbed feet of duck

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15. Which one of the following ideas was not part of Charles Darwin's theory of evolution by natural selection? (nts 2014)

A. Organisms produce more offspring than the environment can support
B. Variation between individuals arises by gene mutation
C. Only those individuals best adapted to the environment survive and reproduce.
D. Individuals compete for scarce resources
E. Adaptive variation is inherited

16. Which of the following men might have explained the auk's loss of the the ability to fly the following hypothesis?
"Since the auk stooped using its wings, the wing became smaller, and this acquired trait was passed on to the offspring (lumhs 2007)

A. Darwin
B. Mendel
C. De varies
D. Lamarck
E.morgan

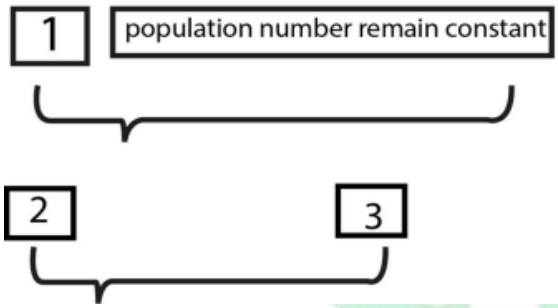
17. There existed two varieties of female fresh water mollusks in which some were streamlined and some had high bulge. Over the generations, male fresh water mollusks learned that high bulge favors more production of offspring. So they started preferring to mate with females having high bulge as compared to streamlined. Will this effect hardy weinberg equilibrium in the population? (lumhs 2019)

A. No, it will not the affect hardy weinberg equilibrium
B. Yes, it will not the affect hardy weinberg equilibrium
C. It will help to balance equilibrium
D. Hardy weinberg equilibrium doesn't apply here

18. What is q in hardy weinberg equilibrium? (nts 2019)

A. Frequency of dominant allele
B. Frequency of the recessive allele
C. Frequency of both alleles
D. Number of both allele

19. The diagram that shows a summary of darwin's theory of natural selection. (pumhs2015)



Natural selection

Which statement should be placed in boxes 1, 2, and 3?

	There is struggle for existence	Variation is shown in all proportions	Individuals show great reproductive capacity
A	1	2	3
B	2	1	3
C	2	3	1
D	3	1	2

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E	3	2	1

20. According to Lamarck, evolution occurred as the result of: (pumhs 2015)
A. Natural selection
B. The theory of the recapitulation
C. Over production
D. Inheritance of acquired characteristics

21. Nictating membrane is found in _____ and performs _____functions(pumhs 2016)
A. Eyes … cleaning
B. Brain .. Thinking
C. Feet .. Locomotion
D. Bones .. Supporting

22. Which of the following statements is false based on darwin's theory of evolution?
A. Natural selection is the driving force of evolution.
B. Favorable genetic variations become more and more common in individuals throughout their lives.
C. There is always a tendency of overproduction in a species.
D. All individuals of a species are not 100% alike.
E. None of the above.

23. Darwin's theory was based on all of following points except: (lumhs 2012)
A. Over reproduction
B. Struggle for existence
C. C. Variations and heredity
D. Survival of the fittest
E. Use and disuse of organs

24. Which one of the terms below correctly describes the relationship between the skeletal system in the wing of bird, fore-limb of turtle, fore-leg of horse and arm of man? (nts smmbbmu 2013)
A. Analogous
B. Homologous
C. Homozygous
D. Monotypic
E. Taxonomic

25. The remains of Archeopteryx indicate that birds are most closely related to
A. Flying insects
B. Flying mammals
C. Flying fish
D. Reptiles
E. Amphibian

26. All of the following are factors in the development of new species except
A. Use and disuse of organ
B. Asexual reproduction
C. Sexual reproduction
D. Variation
E. Natural selection

27. Genetic equilibrium is a: (LUMHS 2019)
A. Change of allele and gene frequency in a population
B. Stability of allele and gene quantity in a population
C. Change of allele and gene number in a Population

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D. Stability of allele and gene frequency in a population

28. In a forest there are a lot of plants, trees shrubs and herbs. What will the palm trees face if they grow in the same forest? (LUMHS 2019)
A. Intra specific competition
B. Inter specific competition
C. Environmental competition
D. All of the above

ANSWER KEYS

1	B	7	C	13	C	19	C	25	D
2	A	8	D	14	B	20	D	26	B
3	D	9	D	15	B	21	A	27	B
4	A	10	A	16	D	22	E	28	D
5	C	11	A	17	D	23	E		
6	A	12	E	18	B	24	B		

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